



Signals and Systems EE252, Spring 2026

Instructor: Tariq Mumtaz

Semester week # 7-8





Course Learning Outcome (CLOs):

CLO-1	Apply knowledge of signals and systems to categorize and perform basic operations on signals and systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-2	Calculate the response of DT and CT LTI systems to various inputs and categorize LTI systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-3	Analyze DT signals and systems in the transform domain using the Z transform.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-4	Analyze CT signals and systems in the transform domain using Laplace, Fourier transforms, and Fourier Series.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-5	Investigate complex engineering problems in signals and systems and formally present the results.	PLO-4 (Investigation)	Cognitive Level-5 (Evaluating)



Need for Transforms in Signal & Systems Analysis:

- Transforms provide frequency-domain insight
- Transforms help to attain a compact mathematical description of Signals & Systems using complex exponentials.
 - Transforms allow decomposition of the signal into complex sinusoids ($e^{j\omega}$)
 - Transforms also allow system representation in exponential form ($e^{\sigma+j\omega}$ or $re^{j\omega}$)
- Transforms convert convolution in the time domain into multiplication in the frequency domain
- Transform converts differential (continuous time)/ difference equations (discrete time) into algebraic equations.



Need for Transforms in Signal & Systems Analysis:

- CTFT/DTFT: decomposition into pure complex exponentials $e^{j\omega t}$ or $e^{j\omega n}$.
- Laplace/Z-Transform: decomposition into general complex exponentials e^{st} or z^n
- CTFT and DTFT are primarily used for spectral (frequency) analysis, while Laplace and Z-transforms are more powerful for system analysis because they include stability/causality and transient behavior.

$$\text{CTFT} = X(s) \Big|_{s=j\omega}$$

$$x(t) \xleftrightarrow{\text{CTFT}} X(\omega)$$

$$x(t) \xleftrightarrow{\text{Laplace Transform}} X(s)$$

$$x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\Omega})$$

$$x[n] \xleftrightarrow{\text{Z Transform}} X(z)$$

$$\text{DTFT} = X(z) \Big|_{z=e^{j\Omega}}$$

Roadmap till Semester end:



Frequency Domain Analysis of CT/DT Signals & Systems:

Z- Transform (Wk 8 – 10)

Laplace Transform (Wk 11 – 13)

Fourier Transform (Wk 14 – 15)

Project presentations (Wk16)

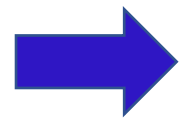


Learning outcome current week sessions:

Z- Transform – 5.1

- Concept & Formulae
- Z and Region of Convergence (ROC)
- Z Transform properties
- Z- Inverse and Partial fraction expansion

Announcements:

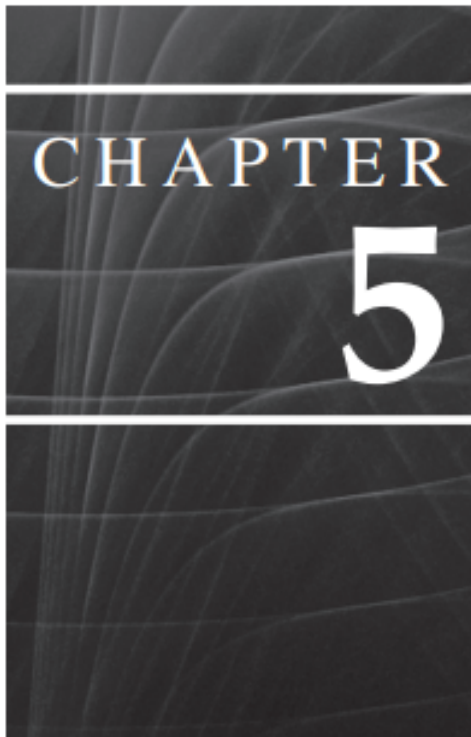


Course Project Announcement (15%)

Midterm Exam (25%) – 9th March, Monday
(Chapters 1 & 3)

The Z Transform

Signal & System Analysis in the Complex Frequency Domain



DISCRETE-TIME SYSTEM ANALYSIS USING THE z -TRANSFORM



Z Transform: Formulae

Transformation of the DT signal from time to the complex frequency domain

Z Transform:

- The Z-transform is a mathematical tool used in signal processing and control systems to convert discrete-time signals into the complex frequency domain.
- The Z-transform helps to represent an arbitrary input $x[n]$ as a sum of (everlasting) exponentials of the form z^n

$$X[z] = \sum_{n=0}^{\infty} x[n]z^{-n} = x[0] + \frac{x[1]}{z} + \frac{x[2]}{z^2} + \frac{x[3]}{z^3} + \dots$$

Z Transform:

$$X[z] = \sum_{n=0}^{\infty} x[n]z^{-n} = x[0] + \frac{x[1]}{z} + \frac{x[2]}{z^2} + \frac{x[3]}{z^3} + \dots$$

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

here z is a complex variable. The signal $x[n]$, which is the inverse z -transform of $X[z]$, can be obtained from $X[z]$ by using the following inverse z -transformation:

$$x[n] = \frac{1}{2\pi j} \oint X[z]z^{n-1} dz$$

That's the integration around a closed path in the complex plane.

Z Transform:

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$x[n]$ = discrete-time signal

z = complex variable

$$z = re^{j\omega}$$

$X(z)$ = Z-transform of the signal

$$X[z] = \mathcal{Z}\{x[n]\} \quad \text{and} \quad x[n] = \mathcal{Z}^{-1}\{X[z]\}$$

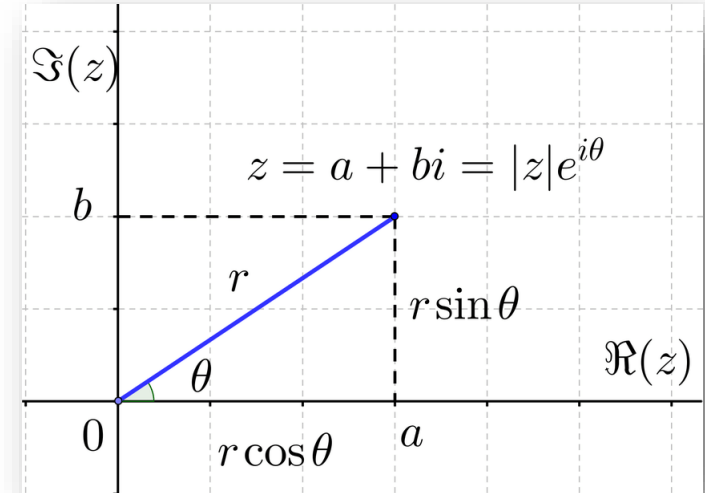
$$x[n] \Longleftrightarrow X[z]$$

Z Transform:

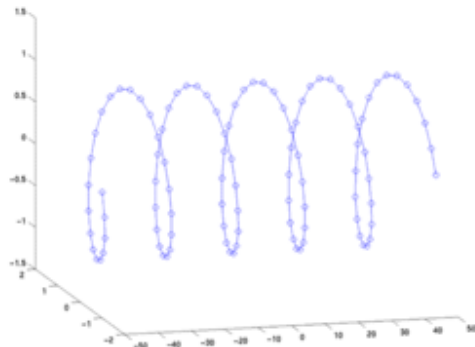
What is Z

- The Z-transform enables frequency domain analysis of discrete-time systems by mapping signals from the time domain to the complex z-plane.
- Z is complex number $Z = a + jb$. While Z^n is a complex exponential

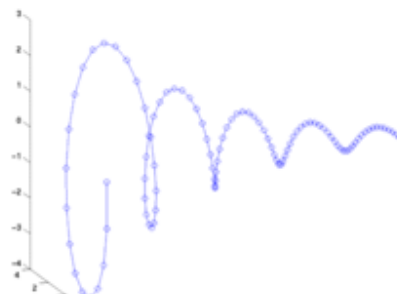
$$z^n = |z|^n e^{j\omega n}$$



$$|z| = 1$$



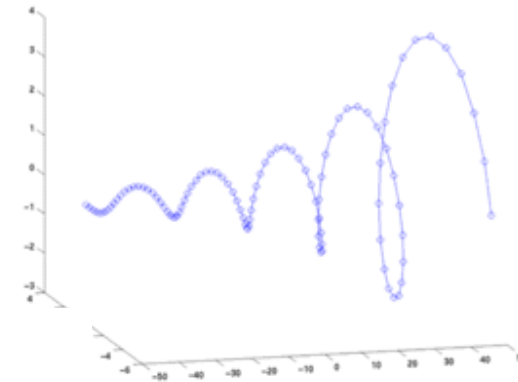
$$|z| < 1$$



z is complex: $z = re^{j\omega}$

Not just frequency — includes growth/decay

$$|z| > 1$$





Z Transform:

$$z = r e^{j\omega}$$

$$z^{-n} = r^{-n} e^{-j\omega n}$$

$$\begin{array}{l} e^{-j\omega n} \rightarrow \text{oscillation} \\ r^{-n} \rightarrow \text{exponential growth or decay} \end{array}$$

The Z-transform expresses a signal $x[n]$ as a weighted (linear) combination of complex exponentials, possibly growing or decaying.

$$x[n] z^{-n}$$

$$x[n] \times (\text{decaying/growing complex exponential})$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Overall, The Z-transform computes how much of each exponentially growing or decaying complex exponential exists in the arbitrary DT signal



Z Transform

Bilateral Z-transform: The two-sided z- transform is used for the $x[n]$ signals over all time

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Unilateral Z-transform: The unilateral Z-transform is used for causal signals $x[n]$

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=0}^{\infty} x[n] z^{-n}$$



Z Transform:

Example: let $x[n] = \delta[n]$ – a delta signal.

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

$$X(z) = \sum_{n=0}^{\infty} \delta[n] z^{-n}$$

Since $\delta[n] = 0$ for $n \neq 0$, the sum reduces to:

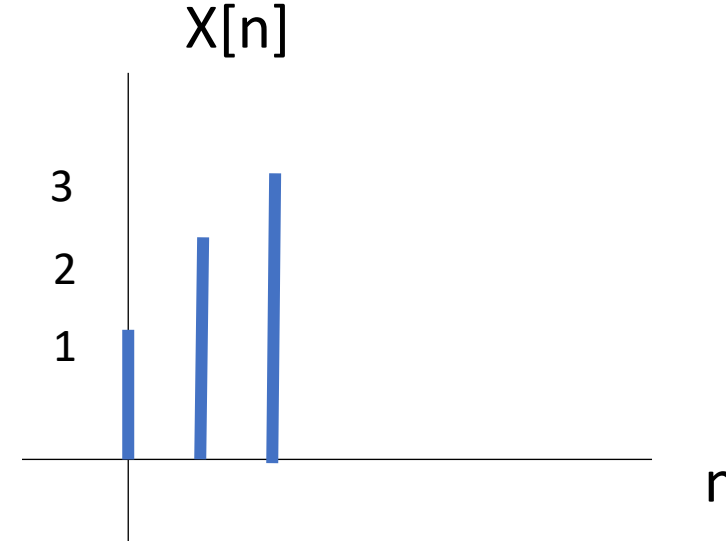
$$X(z) = \delta[0] z^0 = 1$$

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Z Transform:

Example: let $x[n] = [1, 2, 3]$.

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$



$$X(Z) = x[0]z^0 + x[1]z^{-1} + x[2]z^{-2} + x[3]z^{-3} + \dots$$

$$X(Z) = 1 z^0 + 2 z^{-1} + 3 z^{-2} + 0 z^{-3} + \dots$$

$$X(Z) = 1 + 2 z^{-1} + 3 z^{-2}$$



Z Transform:

Example: let $x[n] = \delta[n-k]$ – a shifted delta signal.

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

$$X(z) = \sum_{n=0}^{\infty} \delta[n-k] z^{-n}$$

$$m = n - k \quad \Rightarrow \quad n = m + k$$

When $n \rightarrow \infty$, $m \rightarrow \infty$

$$X(z) = \sum_{m=-k}^{\infty} \delta[m] z^{-(m+k)}$$

$$X(z) = z^{-(0+k)} = z^{-k}$$



Z Transform:

Example: let $x[n] = u[n]$

$$u[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \dots = \sum_{k=0}^{\infty} \delta[n-k]$$

$$X(z) = \sum_{k=0}^{\infty} z^{-k} = 1 + z^{-1} + z^{-2} + \dots$$

Geometric Series

$$X(z) = \sum_{n=0}^{\infty} z^{-n}$$

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}, \quad |r| < 1$$

$$r = z^{-1} \Rightarrow |z^{-1}| < 1 \Rightarrow |z| > 1$$

$$X(z) = \frac{1}{1-z^{-1}}, \quad |z| > 1$$

$$X(z) = \frac{1}{1-z^{-1}} = \frac{z}{z-1},$$

Z Transform:

Example: $x[n] = \gamma^n u[n]$

$$X[z] = \sum_{n=0}^{\infty} \gamma^n u[n] z^{-n}$$

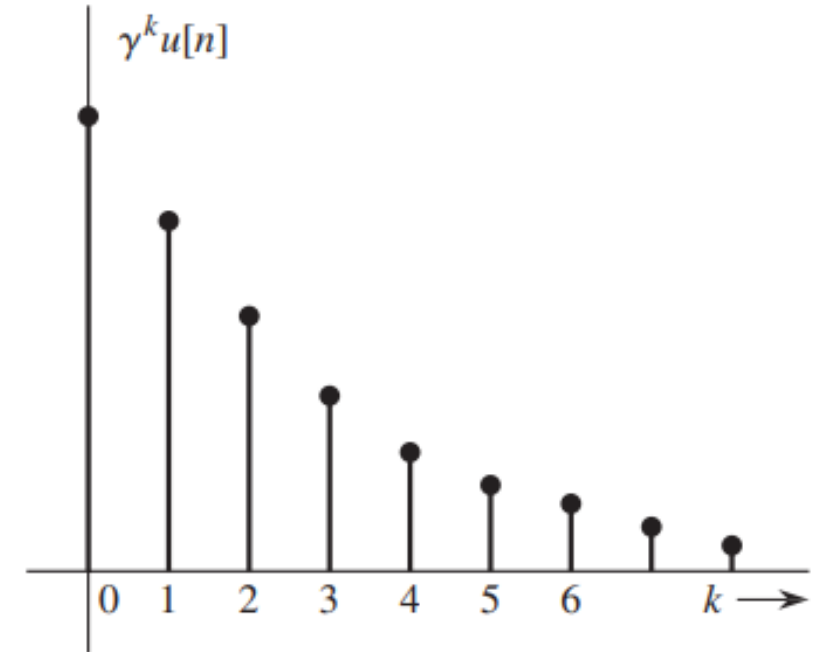
Since $u[n] = 1$ for all $n \geq 0$,

$$X[z] = \sum_{n=0}^{\infty} \left(\frac{\gamma}{z}\right)^n = 1 + \left(\frac{\gamma}{z}\right) + \left(\frac{\gamma}{z}\right)^2 + \left(\frac{\gamma}{z}\right)^3 + \dots + \dots$$

$$1 + x + x^2 + x^3 + \dots = \frac{1}{1-x} \quad \text{if } |x| < 1$$

Applying this relationship to Eq. (5.4) yields

$$\begin{aligned} X[z] &= \frac{1}{1 - \frac{\gamma}{z}} \quad \left| \frac{\gamma}{z} \right| < 1 \\ &= \frac{z}{z - \gamma} \quad |z| > |\gamma| \end{aligned}$$



$$X(z) = \frac{z}{z - \gamma} \quad |z| > |\gamma|$$



Z Transform:

TABLE 5.1 Select (Unilateral) z -Transform Pairs

No.	$x[n]$	$X[z]$
1	$\delta[n - k]$	z^{-k}
2	$u[n]$	$\frac{z}{z - 1}$
3	$nu[n]$	$\frac{z}{(z - 1)^2}$
4	$n^2u[n]$	$\frac{z(z + 1)}{(z - 1)^3}$
5	$n^3u[n]$	$\frac{z(z^2 + 4z + 1)}{(z - 1)^4}$
6	$\gamma^n u[n]$	$\frac{z}{z - \gamma}$
7	$\gamma^{n-1} u[n - 1]$	$\frac{1}{z - \gamma}$
8	$n\gamma^n u[n]$	$\frac{\gamma z}{(z - \gamma)^2}$
9	$n^2\gamma^n u[n]$	$\frac{\gamma z(z + \gamma)}{(z - \gamma)^3}$

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$



Z Transform:

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n] z^{-n} \quad 10$$

$$\frac{n(n-1)(n-2) \cdots (n-m+1)}{\gamma^m m!} \gamma^n u[n] \quad 10$$

$$|\gamma|^n \cos \beta n u[n] \quad 11a$$

$$|\gamma|^n \sin \beta n u[n] \quad 11b$$

$$r |\gamma|^n \cos (\beta n + \theta) u[n] \quad 12a$$

$$r |\gamma|^n \cos (\beta n + \theta) u[n] \quad \gamma = |\gamma| e^{j\beta} \quad 12b$$

$$r |\gamma|^n \cos (\beta n + \theta) u[n] \quad 12c$$

$$r = \sqrt{\frac{A^2 |\gamma|^2 + B^2 - 2AaB}{|\gamma|^2 - a^2}}$$

$$\beta = \cos^{-1} \frac{-a}{|\gamma|}$$

$$\theta = \tan^{-1} \frac{Aa - B}{A\sqrt{|\gamma|^2 - a^2}}$$

$$\frac{z}{(z - \gamma)^{m+1}}$$

$$\frac{z(z - |\gamma| \cos \beta)}{z^2 - (2|\gamma| \cos \beta)z + |\gamma|^2}$$

$$\frac{z|\gamma| \sin \beta}{z^2 - (2|\gamma| \cos \beta)z + |\gamma|^2}$$

$$\frac{rz[z \cos \theta - |\gamma| \cos (\beta - \theta)]}{z^2 - (2|\gamma| \cos \beta)z + |\gamma|^2}$$

$$\frac{(0.5re^{j\theta})z}{z - \gamma} + \frac{(0.5re^{-j\theta})z}{z - \gamma^*}$$

$$\frac{z(Az + B)}{z^2 + 2az + |\gamma|^2}$$

End of session WK7- Session 02



Z Transform: Signals Perspective - ROC



Z Transform:

The sum in the expression of the z-transform may not converge (exist) for all values of z .

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

The values of z (the region in the complex plane) for which the sum converges (or exists) are called the region of convergence (ROC), for $X[z]$.

$$X[z] = \sum_{n=0}^{\infty} x[n]z^{-n} = x[0] + \frac{x[1]}{z} + \frac{x[2]}{z^2} + \frac{x[3]}{z^3} + \dots$$

Z Transform:

Example: $x[n] = \gamma^n u[n]$

$$X[z] = \sum_{n=0}^{\infty} \gamma^n u[n] z^{-n}$$

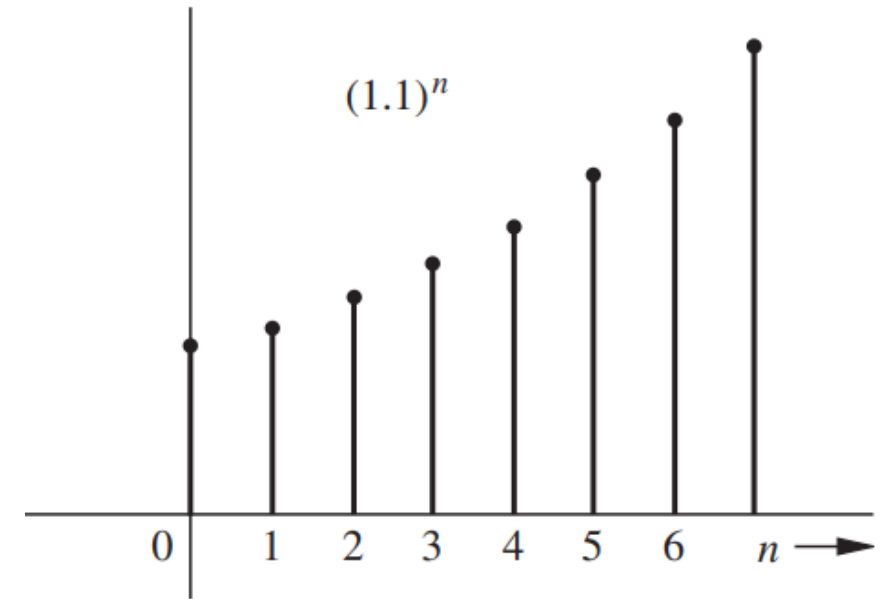
Since $u[n] = 1$ for all $n \geq 0$,

$$X[z] = \sum_{n=0}^{\infty} \left(\frac{\gamma}{z}\right)^n = 1 + \left(\frac{\gamma}{z}\right) + \left(\frac{\gamma}{z}\right)^2 + \left(\frac{\gamma}{z}\right)^3 + \dots + \dots$$

$$1 + x + x^2 + x^3 + \dots = \frac{1}{1-x} \quad \text{if } |x| < 1$$

Applying this relationship to Eq. (5.4) yields

$$\begin{aligned} X[z] &= \frac{1}{1 - \frac{\gamma}{z}} \quad \left| \frac{\gamma}{z} \right| < 1 \\ &= \frac{z}{z - \gamma} \quad |z| > |\gamma| \end{aligned}$$



$$X(z) = \frac{z}{z - \gamma} \quad |z| > |\gamma|$$

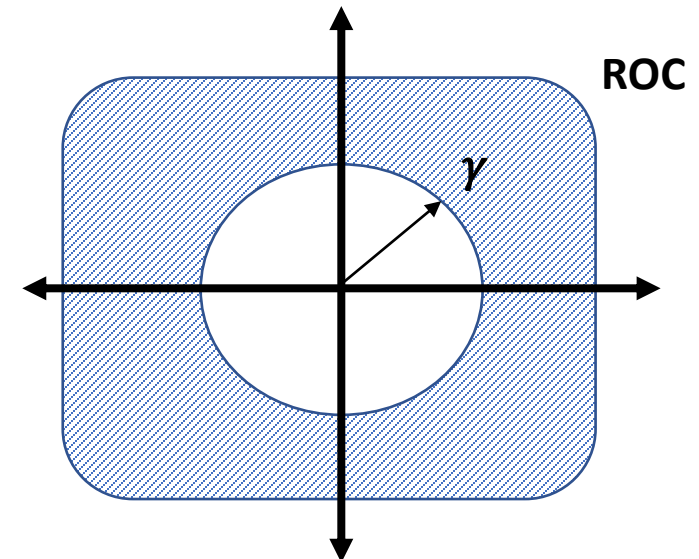
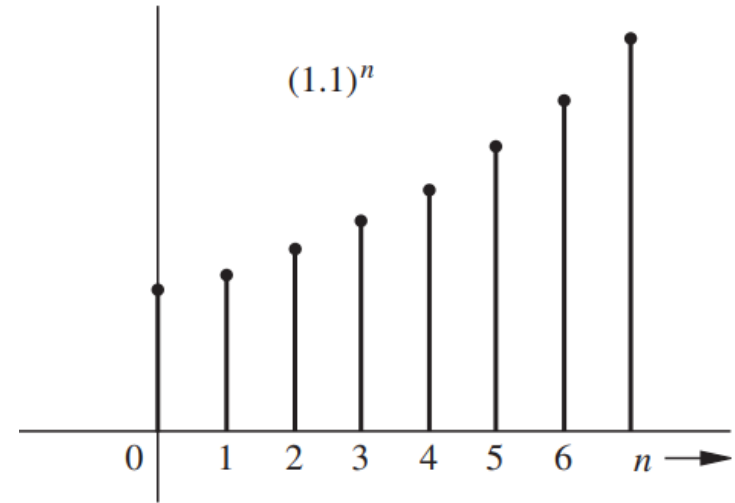
Z Transform:

$$X(z) = \frac{z}{z - \gamma} \quad |z| > |\gamma|$$

If the signal grows like γ^n , then the weighting z^{-n} must decay faster to ensure convergence.

$$X[z] = \sum_{n=0}^{\infty} x[n]z^{-n} = x[0] + \frac{x[1]}{z} + \frac{x[2]}{z^2} + \frac{x[3]}{z^3} + \dots$$

ROC lies outside the circle $|z| = |\gamma|$.





Z Transform:

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

For $x[n] = \gamma^n u[n]$

$$\gamma^n \times z^{-n}$$

For convergence

$$\left| \frac{\gamma}{z} \right| < 1$$

$$|z| > |\gamma|$$

For convergence, if the signal grows, the weighting z^{-n} must decay faster.

ROC lies outside the circle $|z| > |\gamma|$

Z Transform:

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

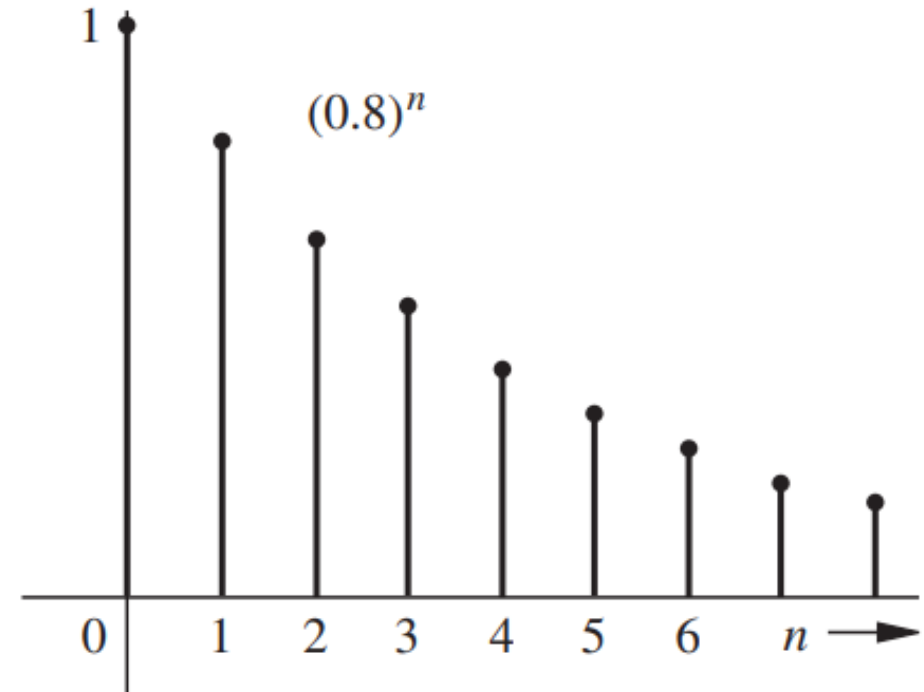
Example: $x[n] = \gamma^n u[n]$ when γ is less than 1

What will be the signal Z transform?

What will be the ROC?

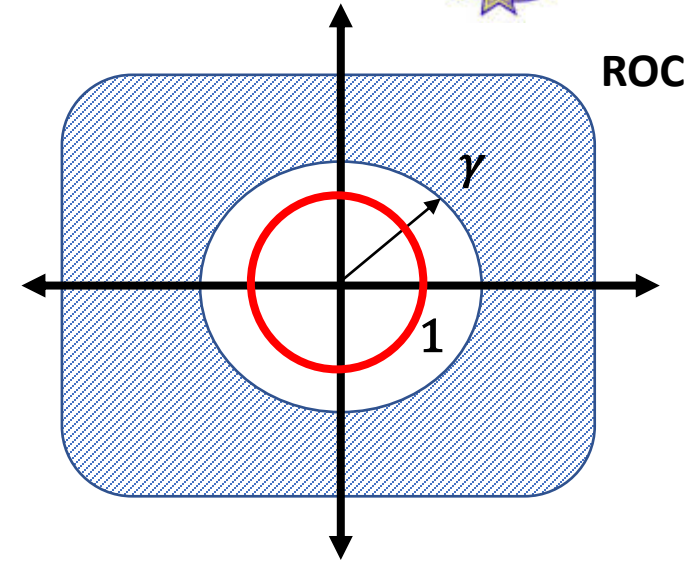
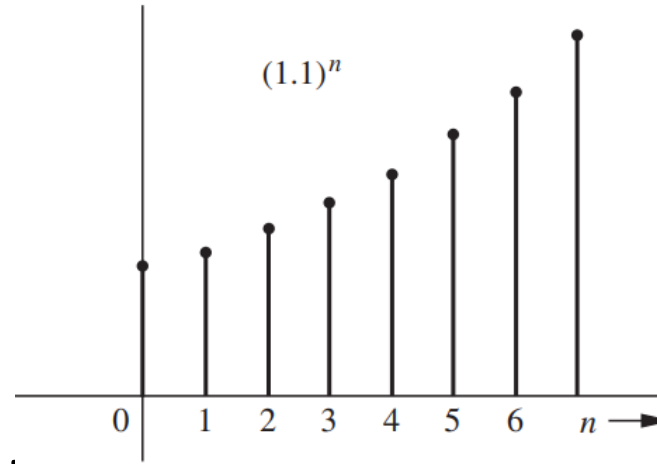
Is the convergence Rule still valid?

For convergence, if the signal grows, the weighting z^{-n} must decay faster.



Z Transform:

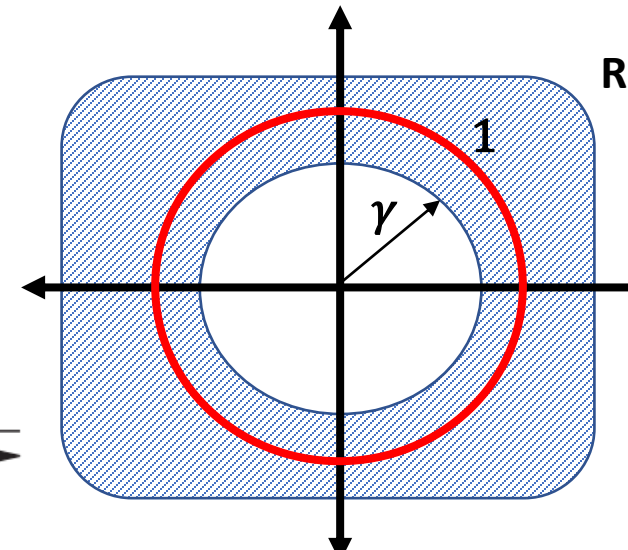
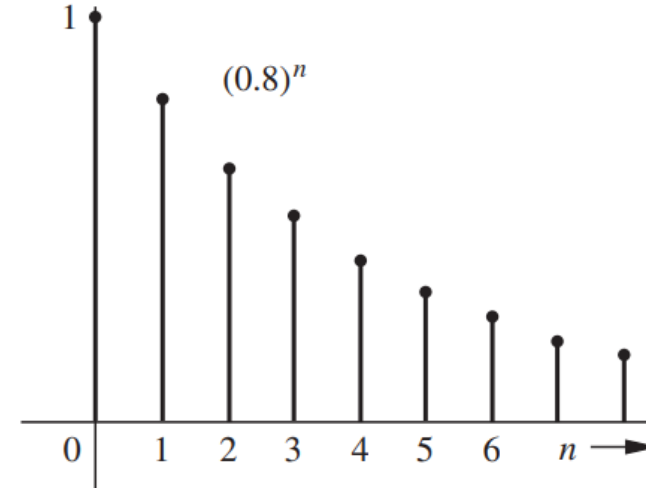
$$X(z) = \frac{z}{z - \gamma} \quad |z| > |\gamma|$$



If the signal decays like γ^n , then the weighting z^{-n} must also decay faster to ensure quick convergence.

$$X[z] = \sum_{n=0}^{\infty} x[n]z^{-n} = x[0] + \frac{x[1]}{z} + \frac{x[2]}{z^2} + \frac{x[3]}{z^3} + \dots$$

↓
↑



ROC lies outside the circle $|z| > |\gamma|$

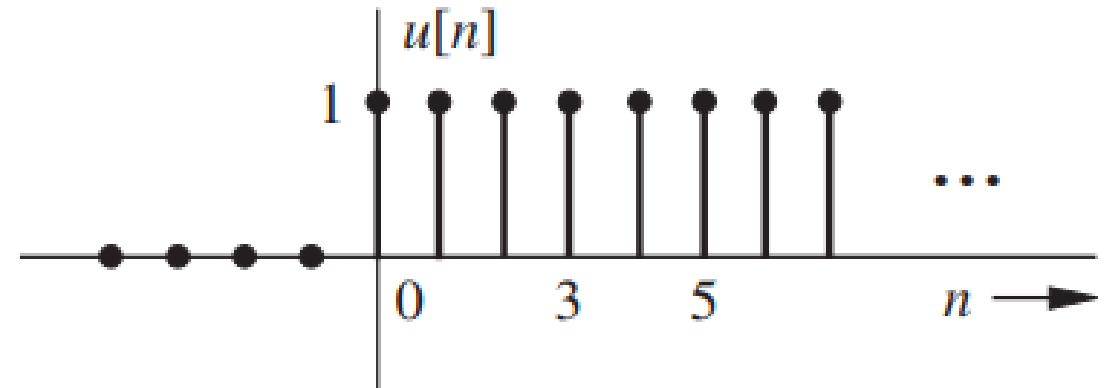
Z Transform:

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Example: $x[n] = \gamma^n u[n]$ when γ is equal to 1 (not a growing signal)

What will be the signal Z transform?

What will be the ROC?



Is the convergence Rule still valid?

For convergence, if the signal grows, the weighting z^{-n} must decay faster

Z Transform:

Example: let $x[n] = u[n]$ – special case of γ^n - here $\gamma = 1$

$$u[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \dots = \sum_{k=0}^{\infty} \delta[n-k]$$

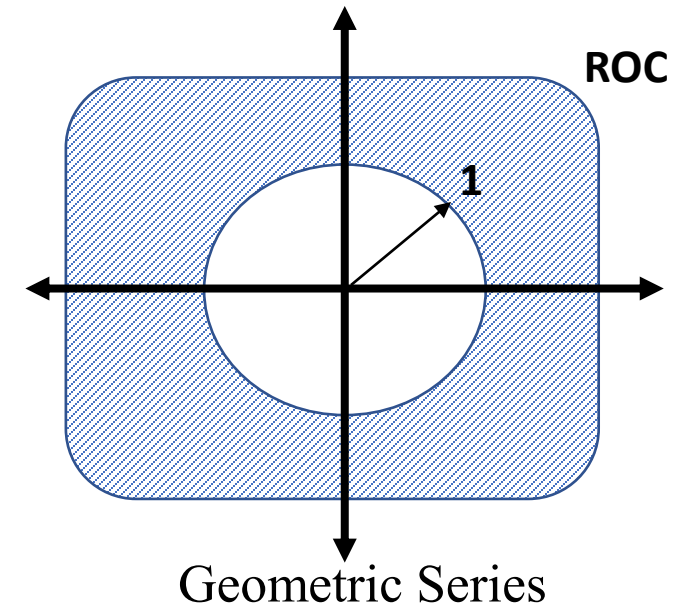
$$X(z) = \sum_{k=0}^{\infty} z^{-k} = 1 + z^{-1} + z^{-2} + \dots$$

$$X(z) = \sum_{n=0}^{\infty} z^{-n}$$

$$r = z^{-1} \Rightarrow |z^{-1}| < 1 \Rightarrow |z| > 1$$

$$X(z) = \frac{1}{1 - z^{-1}}, \quad |z| > 1$$

$$X(z) = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1}, \quad \text{ROC: } |z| > 1$$



$$\sum_{n=0}^{\infty} r^n = \frac{1}{1 - r}, \quad |r| < 1$$

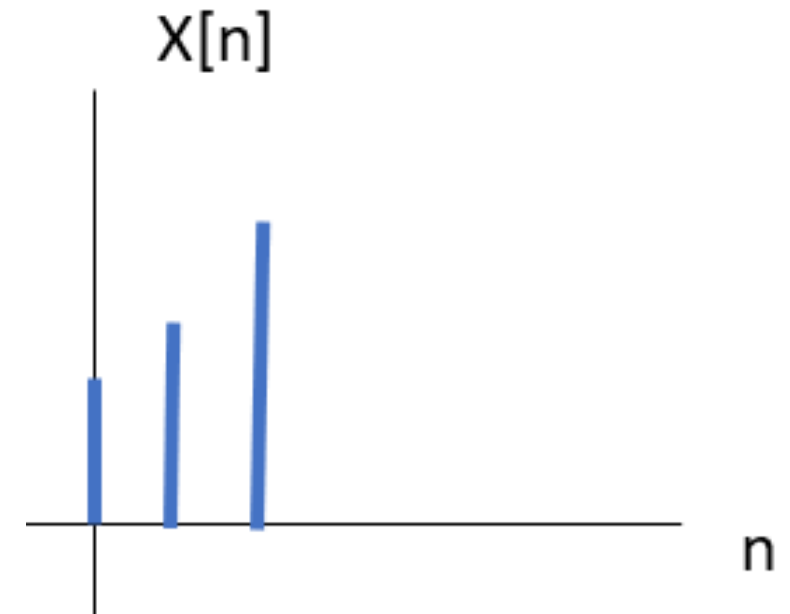
Z Transform:

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Example: $x[n] = [1, 2, 3]$ - a growing signal but with finite length

What will be the signal Z transform?

What will be the ROC?



Z Transform:

Example: let $x[n] = [1, 2, 3]$.

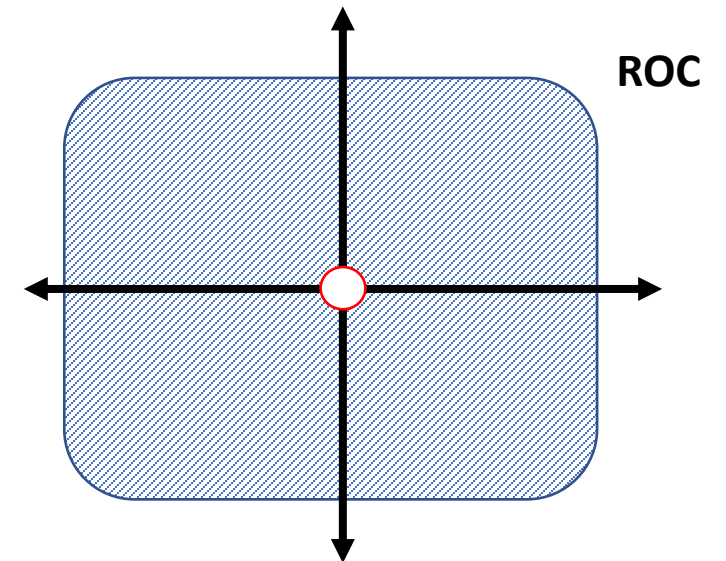
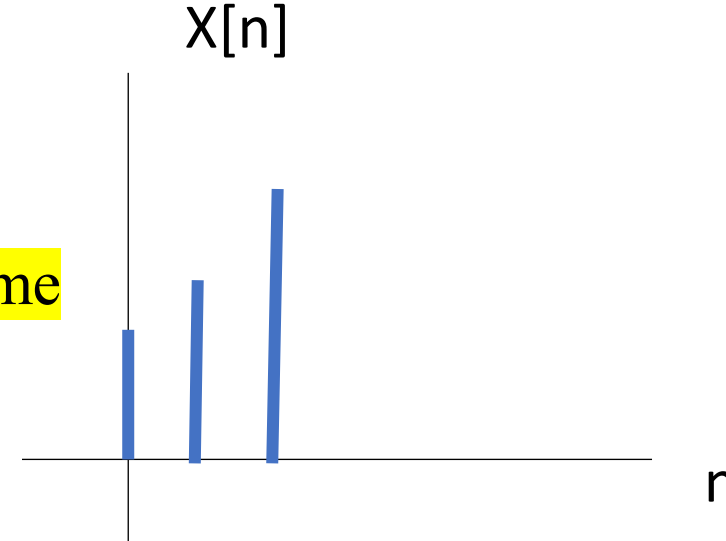
Finite DT signal, not growing to infinity with time

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$

$$X(Z) = x[0]z^0 + x[1]z^{-1} + x[2]z^{-2} + x[3]z^{-3} + \dots$$

$$X(Z) = 1 z^0 + 2 z^{-1} + 3 z^{-2} + 0 z^{-3} + \dots$$

$$X(Z) = 1 + 2 z^{-1} + 3 z^{-2}$$



There is no infinite summation here; it's just a few terms. So convergence is not about a series.

ROC: entire z-plane except $z=0$

The center $z=0$ corresponds to extreme decay

Z Transform:

Example: let $x[n] = \delta[n]$ – a delta function.

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

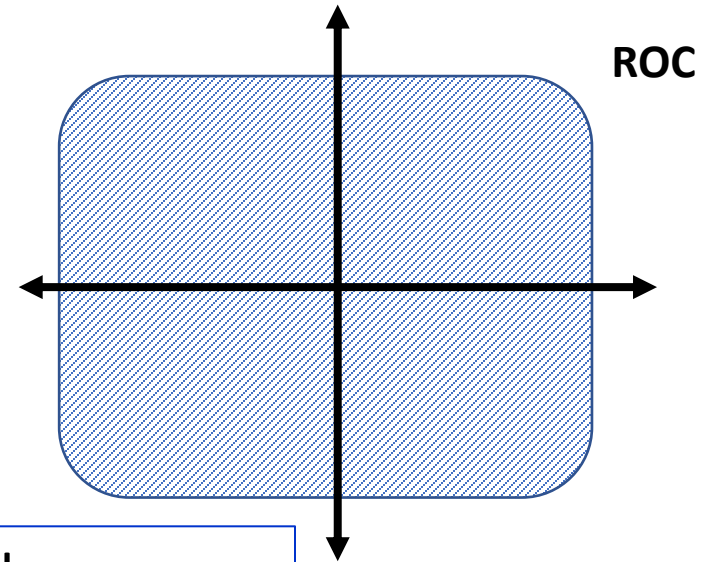
$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n] z^{-n}$$

Since $\delta[n] = 0$ for $n \neq 0$, the sum reduces to:

$$X(z) = \delta[0] z^0 = 1$$

ROC: Entire Z-plane



Since the delta signal exists only at $n = 0$, it is completely unaffected by any exponential weighting z^{-n} , and every point in the Z-plane produces the same response. Therefore, the Z-transform of $\delta[n]$ is constant everywhere, and its ROC is the entire Z-plane.



Z Transform:

Example: let $x[n] = \delta[n - k]$ – a shifted delta function.

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

$$X(z) = \sum_{n=0}^{\infty} \delta[n - k] z^{-n}$$

$$m = n - k \quad \Rightarrow \quad n = m + k$$

When $n \rightarrow \infty$, $m \rightarrow \infty$

$$X(z) = \sum_{m=-k}^{\infty} \delta[m] z^{-(m+k)}$$

$$X(z) = z^{-(0+k)} = z^{-k}$$

There is no infinite summation here; it's just a single term. So convergence is not about a series.

Z Transform:

Example: let $x[n] = \delta[n - k]$ – a shifted delta function.

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n - k] z^{-n}$$

Since the delta is nonzero only at $n=k$

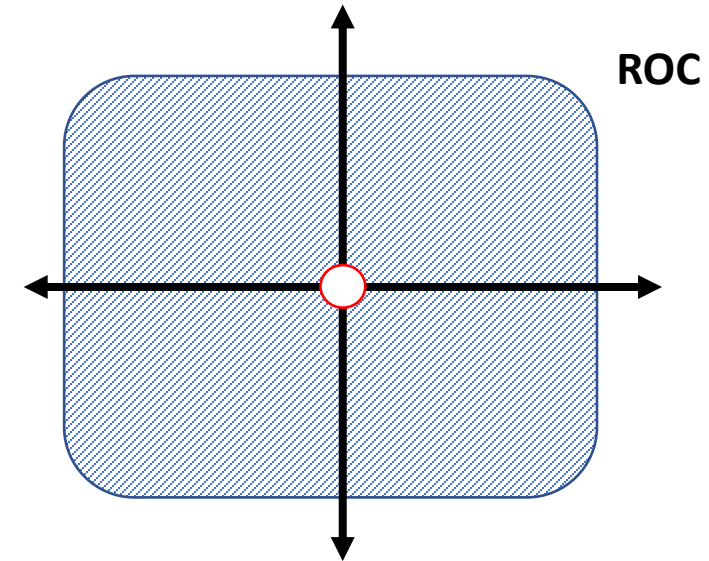
$$X(z) = z^{-k}$$

Case # 01 - When $K > 0$ - left shift

$$X(z) = \frac{1}{z^k}$$

$X(Z)$ becomes infinite at

$$z = 0$$



ROC: entire z-plane except $z=0$

Z Transform:

Example: let $x[n] = \delta[n - k]$ – a shifted delta function.

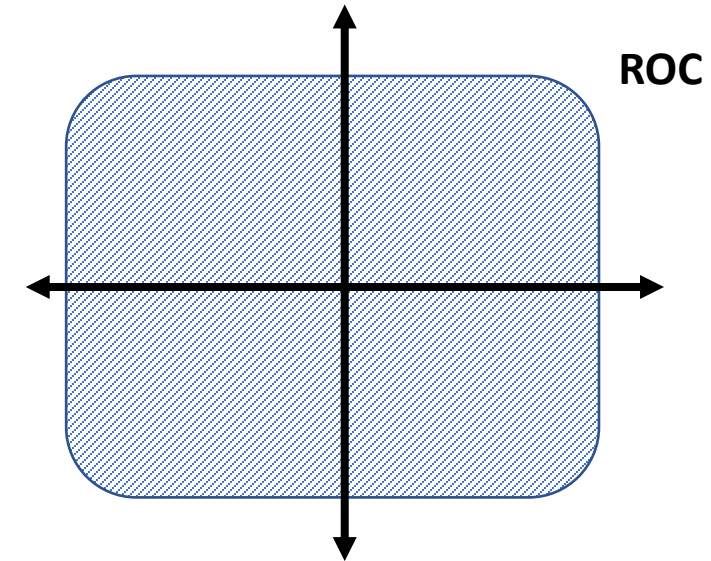
$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n - k] z^{-n}$$

Since the delta is nonzero only at $n=k$

$$X(z) = z^{-k}$$

Case # 02 - When $K = 0$ - No shift

$$X(z) = \delta[0] z^0 = 1$$



ROC: Entire Z-plane

Z Transform:

Example: let $x[n] = \delta[n - k]$ – a shifted delta function.

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n - k] z^{-n}$$

Since the delta is nonzero only at $n=k$

$$X(z) = z^{-k}$$

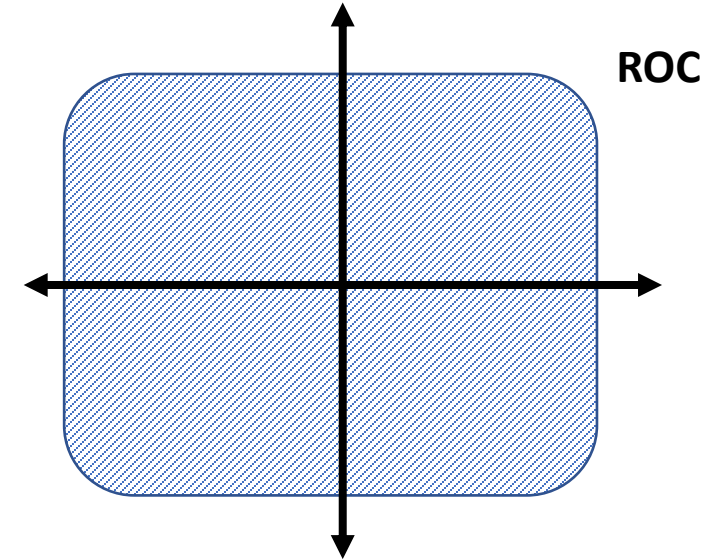
Case # 03 - When $K < 0$ - Right shift

$$X(Z) = Z^K$$

$X(Z)$ becomes infinite at

$$Z = \infty$$

Practically all finite z



ROC: entire z -plane except $z = \infty$

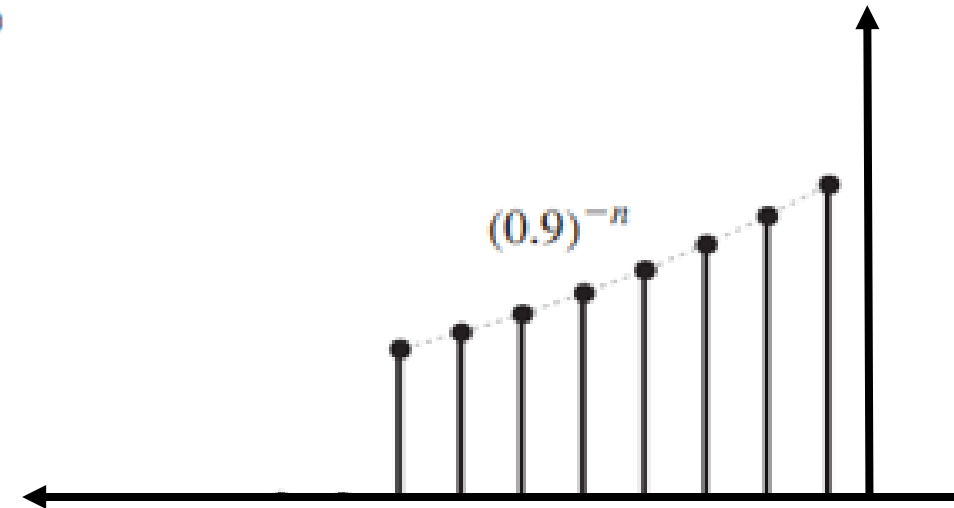
Z Transform:

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

For $x[n] = \gamma^n u[-n-1]$

What will be the signal Z transform?

What will be the ROC?



left-sided signal, exists in negative time only

Is the convergence Rule still valid?

For convergence, if the signal grows, the weighting z^{-n} must decay faster.

Z Transform:

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

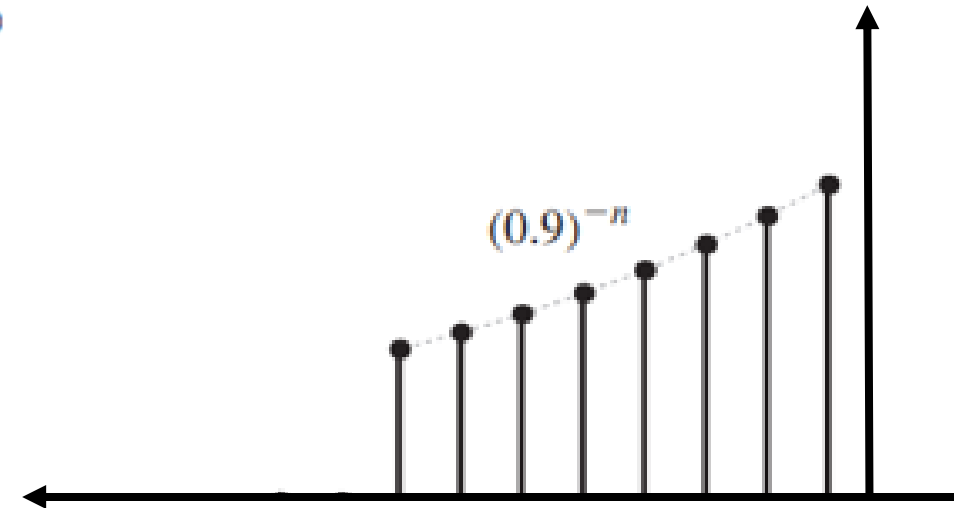
For $x[n] = \gamma^n u[-n-1]$

$$X(z) = \sum_{n=-\infty}^{-1} \gamma^n z^{-n}$$

$$X(z) = \sum_{n=-\infty}^{-1} (\gamma z^{-1})^n$$

$$X(Z) = \left(\frac{\gamma}{z}\right)^{-1} + \left(\frac{\gamma}{z}\right)^{-2} + \left(\frac{\gamma}{z}\right)^{-3} \dots \dots \dots$$

$$X(Z) = \left(\frac{z}{\gamma}\right) + \left(\frac{z}{\gamma}\right)^2 + \left(\frac{z}{\gamma}\right)^3 \dots \dots \dots$$



$$X(z) = \frac{z}{\gamma} + \frac{z^2}{\gamma^2} + \frac{z^3}{\gamma^3} + \frac{z^4}{\gamma^4} + \dots$$

For convergence $\left| \frac{z}{\gamma} \right| < 1 \implies |z| < |\gamma|$

Z Transform:

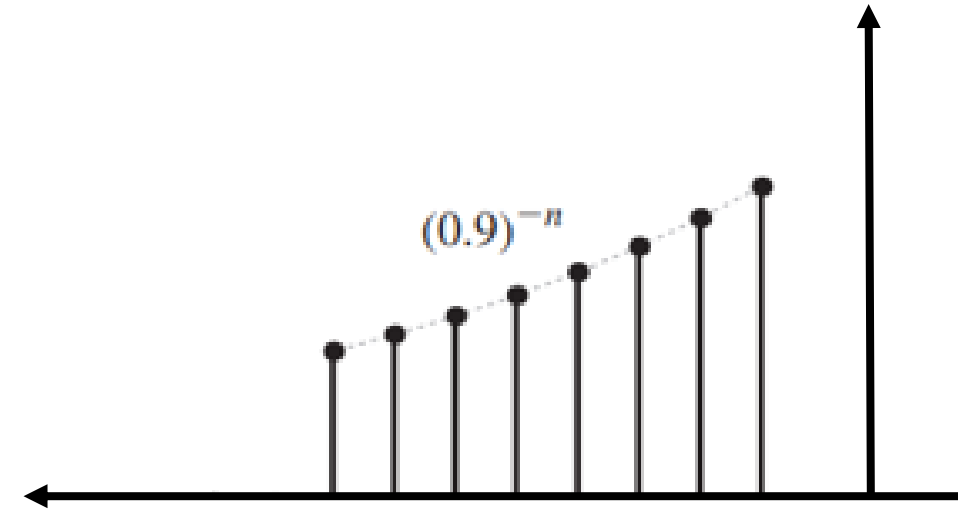
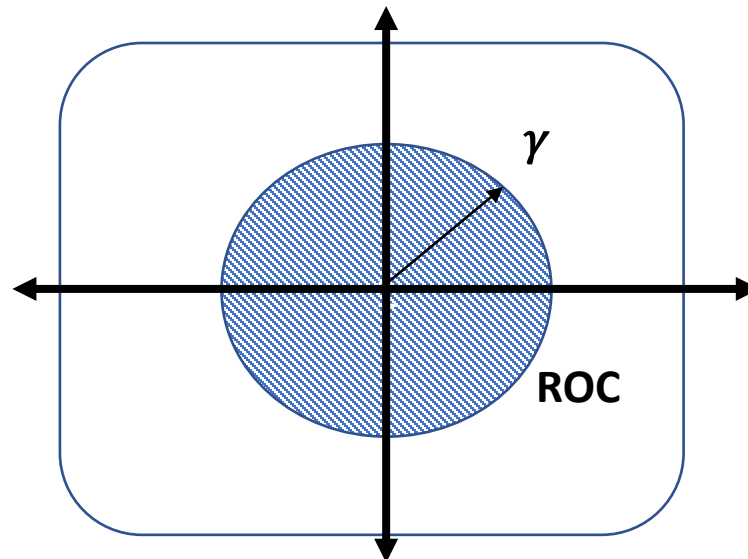
$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

For $x[n] = \gamma^n u[-n-1]$

$$X(z) = \frac{z}{\gamma} + \frac{z^2}{\gamma^2} + \frac{z^3}{\gamma^3} + \frac{z^4}{\gamma^4} + \dots$$

$$\left| \frac{z}{\gamma} \right| < 1 \implies |z| < |\gamma|$$

ROC: $|z| < |\gamma|$





Z Transform:

$$X(z) = \sum_{n=-\infty}^{-1} (\gamma z^{-1})^n$$

For $x[n] = \gamma^n u[-n-1]$

$$m = -n - 1$$

$$n = -1 \Rightarrow m = 0, n \rightarrow -\infty \Rightarrow m \rightarrow \infty$$

$$X(z) = \sum_{m=0}^{\infty} (\gamma z^{-1})^{-m-1}$$

$$X(z) = \sum_{m=0}^{\infty} \frac{z}{\gamma} \left(\frac{z}{\gamma}\right)^m = \frac{z}{\gamma} \sum_{m=0}^{\infty} \left(\frac{z}{\gamma}\right)^m$$

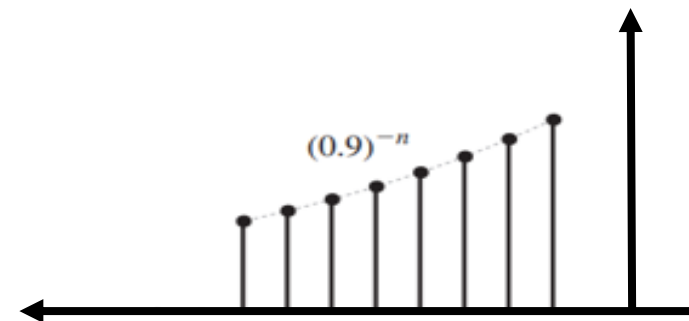
$$X(z) = \frac{z}{\gamma} \cdot \frac{1}{1 - z/\gamma} = \frac{z}{\gamma - z}$$

$$\sum_{m=0}^{\infty} r^m = \frac{1}{1-r}, \quad |r| < 1$$

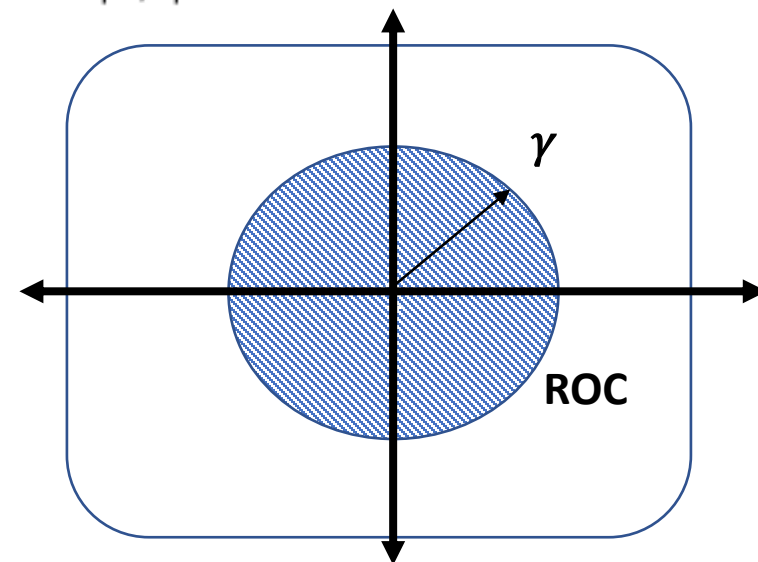
$$r = \frac{z}{\gamma}$$

$$\left|\frac{z}{\gamma}\right| < 1 \Rightarrow |z| < |\gamma|$$

$$\text{ROC: } |z| < |\gamma|$$



$$\left|\frac{z}{\gamma}\right| < 1 \Rightarrow |z| < |\gamma|$$



The Z Transform Properties



Z Transform Properties:

Linearity

$$x[n] = x_1[n] + x_2[n]$$

$$X(z) = Z\{x_1[n]\} + Z\{x_2[n]\}$$

$$Z\{ax_1[n] + bx_2[n]\} = aX_1(z) + bX_2(z)$$

ROC

The new ROC will be the intersection of the ROC of $X_1(z)$ and $X_2(z)$



Z Transform Properties:

Time Reversal

$$x[n] \Longleftrightarrow X[z]$$

$$x[-n] \Longleftrightarrow X[1/z]$$

ROC

The region of convergence is also inverted; that is, if the ROC of $x[n]$ is $|z| > |\gamma|$, then the ROC of $x[-n]$ is $|z| < 1/|\gamma|$.

Z Transform Properties:

Time Shifting

$$x[n]u[n] \iff X[z]$$

$$x[n-1]u[n-1] \iff \frac{1}{z}X[z]$$

ROC

$$x[n-m]u[n-m] \iff \frac{1}{z^m}X[z]$$

The new ROC will be ROC of $X(z)$ plus or minus $\{z=0 \text{ or } z=\infty\}$, depending upon the sign of m



Z Transform Properties:

Time Convolution

$$x_1[n] \iff X_1[z] \quad \text{and} \quad x_2[n] \iff X_2[z],$$

$$x_1[n] * x_2[n] \iff X_1[z]X_2[z]$$

Application of this property in an LTID (Linear Time Invariant DT System)

$$y[n] = x[n] * h[n] \quad \quad \quad h[n] \iff H[z] \quad \quad \quad Y[z] = X[z]H[z]$$

ROC

Depends on whether the result is causal, anti-causal, or two-sided



Z Transform Properties:

Time domain multiplication by a factor γ^n

Multiplying a time-domain signal by an exponential γ^n is equivalent to scaling in the z-domain

$$x[n]u[n] \Longleftrightarrow X[z]$$

$$\gamma^n x[n]u[n] \Longleftrightarrow X\left[\frac{z}{\gamma}\right]$$

ROC

The ROC is scaled by $|\gamma|$ factor



Z Transform Properties:

Time domain multiplication by a factor n

Multiplying a signal by n in the time domain produces differentiation in the z -domain.

$$x[n]u[n] \Longleftrightarrow X[z]$$

$$nx[n]u[n] \Longleftrightarrow -z \frac{d}{dz} X[z]$$

ROC

Multiplying by n does NOT change the ROC



Z Transform Properties:

TABLE 5.2 z -Transform Properties

Please refer to the textbook for proof

Operation	$x[n]$	$X[z]$
Addition	$x_1[n] + x_2[n]$	$X_1[z] + X_2[z]$
Scalar multiplication	$ax[n]$	$aX[z]$
Right shifting	$x[n - m]u[n - m]$	$\frac{1}{z^m}X[z]$
	$x[n - m]u[n]$	$\frac{1}{z^m}X[z] + \frac{1}{z^m} \sum_{n=1}^m x[-n]z^n$
	$x[n - 1]u[n]$	$\frac{1}{z}X[z] + x[-1]$
	$x[n - 2]u[n]$	$\frac{1}{z^2}X[z] + \frac{1}{z}x[-1] + x[-2]$
	$x[n - 3]u[n]$	$\frac{1}{z^3}X[z] + \frac{1}{z^2}x[-1] + \frac{1}{z}x[-2] + x[-3]$



Z Transform Properties:

Left shifting

$$x[n+m]u[n]$$

$$z^m X[z] - z^m \sum_{n=0}^{m-1} x[n]z^{-n}$$

$$x[n+1]u[n]$$

$$zX[z] - zx[0]$$

$$x[n+2]u[n]$$

$$z^2 X[z] - z^2 x[0] - zx[1]$$

$$x[n+3]u[n]$$

$$z^3 X[z] - z^3 x[0] - z^2 x[1] - zx[2]$$

Multiplication by γ^n

$$\gamma^n x[n]u[n]$$

$$X\left[\frac{z}{\gamma}\right]$$

Multiplication by n

$$nx[n]u[n]$$

$$-z \frac{d}{dz} X[z]$$

Time reversal

$$x[-n]$$

$$X[1/z]$$

Time convolution

$$x_1[n] * x_2[n]$$

$$X_1[z]X_2[z]$$

Initial value

$$x[0]$$

$$\lim_{z \rightarrow \infty} X[z]$$

Final value

$$\lim_{N \rightarrow \infty} x[N]$$

$$\lim_{z \rightarrow 1} (z-1)X[z]$$

Poles of $(z-1)X[z]$
inside the unit circle



Z Transform Properties:

Practice of Z- Transform's properties

$$x[n] = 5^{n+1}u[n]$$

$$x[n] = 5 \cdot 5^n u[n]$$

Term: $\gamma^n u[n] \longleftrightarrow \frac{z}{z - \gamma}$

$$X(z) = \frac{5z}{z - 5}$$

ROC $|z| > 5$

TABLE 5.1 Select (Unilateral) z-Transform Pairs

No.	$x[n]$	$X[z]$
1	$\delta[n - k]$	z^{-k}
2	$u[n]$	$\frac{z}{z - 1}$
3	$nu[n]$	$\frac{z}{(z - 1)^2}$
4	$n^2u[n]$	$\frac{z(z + 1)}{(z - 1)^3}$
5	$n^3u[n]$	$\frac{z(z^2 + 4z + 1)}{(z - 1)^4}$
6	$\gamma^n u[n]$	$\frac{z}{z - \gamma}$
7	$\gamma^{n-1} u[n - 1]$	$\frac{1}{z - \gamma}$
8	$n\gamma^n u[n]$	$\frac{\gamma z}{(z - \gamma)^2}$
9	$n^2\gamma^n u[n]$	$\frac{\gamma z(z + \gamma)}{(z - \gamma)^3}$



Z Transform Properties:

Practice of Z- Transform's properties

$$x[n] = (5 + 5^n)u[n]$$

Apply Linearity property:

$$x[n] = 5u[n] + 5^n u[n]$$

$$X(z) = Z\{5u[n]\} + Z\{5^n u[n]\}$$

First Term: $u[n] \longleftrightarrow \frac{z}{z-1}$

Second Term: $\gamma^n u[n] \longleftrightarrow \frac{z}{z-\gamma}$

$$X(z) = \frac{5z}{z-1} + \frac{z}{z-5}$$

- ROC of first term: $|z| > 1$
- ROC of second term: $|z| > 5$
- Overall ROC is the intersection: $|z| > 5$

TABLE 5.1 Select (Unilateral) z-Transform Pairs

No.	$x[n]$	$X[z]$
1	$\delta[n-k]$	z^{-k}
2	$u[n]$	$\frac{z}{z-1}$
3	$nu[n]$	$\frac{z}{(z-1)^2}$
4	$n^2u[n]$	$\frac{z(z+1)}{(z-1)^3}$
5	$n^3u[n]$	$\frac{z(z^2+4z+1)}{(z-1)^4}$
6	$\gamma^n u[n]$	$\frac{z}{z-\gamma}$
7	$\gamma^{n-1} u[n-1]$	$\frac{1}{z-\gamma}$
8	$n\gamma^n u[n]$	$\frac{\gamma z}{(z-\gamma)^2}$
9	$n^2\gamma^n u[n]$	$\frac{\gamma z(z+\gamma)}{(z-\gamma)^3}$



Z Transform Properties:

Practice of Z- Transform's properties

$$x[n] = -2\delta[n - 4] + \left\{ \frac{3}{2}(2)^n + \frac{5}{3}(3)^n \right\} u[n]$$

Using the linearity property:

$$x_1[n] = -2\delta[n - 4]$$

$$x_2[n] = \frac{3}{2}(2)^n u[n]$$

$$x_3[n] = \frac{5}{3}(3)^n u[n]$$

$$X(z) = X_1(z) + X_2(z) + X_3(z)$$

TABLE 5.1 Select (Unilateral) z -Transform Pairs

No.	$x[n]$	$X[z]$
1	$\delta[n - k]$	z^{-k}
2	$u[n]$	$\frac{z}{z - 1}$
3	$nu[n]$	$\frac{z}{(z - 1)^2}$
4	$n^2u[n]$	$\frac{z(z + 1)}{(z - 1)^3}$
5	$n^3u[n]$	$\frac{z(z^2 + 4z + 1)}{(z - 1)^4}$
6	$\gamma^n u[n]$	$\frac{z}{z - \gamma}$
7	$\gamma^{n-1} u[n - 1]$	$\frac{1}{z - \gamma}$
8	$n\gamma^n u[n]$	$\frac{\gamma z}{(z - \gamma)^2}$
9	$n^2\gamma^n u[n]$	$\frac{\gamma z(z + \gamma)}{(z - \gamma)^3}$



Z Transform Properties:

Practice of Z- Transform's properties

$$x[n] = -2\delta[n - 4] + \left\{ \frac{3}{2}(2)^n + \frac{5}{3}(3)^n \right\} u[n]$$

First term: $\delta[n - k] \longleftrightarrow z^{-k}$

2nd and 3rd term:

$$\gamma^n u[n] \longleftrightarrow \frac{z}{z - \gamma}$$

$$X(z) = -2z^{-4} + \frac{3z}{2(z - 2)} + \frac{5z}{3(z - 3)}$$

ROC $|z| > 3$

TABLE 5.1 Select (Unilateral) z-Transform Pairs

No.	$x[n]$	$X[z]$
1	$\delta[n - k]$	z^{-k}
2	$u[n]$	$\frac{z}{z - 1}$
3	$nu[n]$	$\frac{z}{(z - 1)^2}$
4	$n^2u[n]$	$\frac{z(z + 1)}{(z - 1)^3}$
5	$n^3u[n]$	$\frac{z(z^2 + 4z + 1)}{(z - 1)^4}$
6	$\gamma^n u[n]$	$\frac{z}{z - \gamma}$
7	$\gamma^{n-1} u[n - 1]$	$\frac{1}{z - \gamma}$
8	$n\gamma^n u[n]$	$\frac{\gamma z}{(z - \gamma)^2}$
9	$n^2\gamma^n u[n]$	$\frac{\gamma z(z + \gamma)}{(z - \gamma)^3}$

End of session WK8- Session 02



Learning outcome current week sessions:

Z- Transform – 5.1 & 5.2

- Concept & Formulae
- Z and Region of Convergence (ROC)
- Z Transform properties
- Z- Inverse and Partial fraction expansion

Z Transform:

The sum in the expression of the z-transform may not converge (exist) for all values of z .

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

The values of z (the region in the complex plane) for which the sum converges (or exists) are called the region of convergence (ROC), for $X[z]$.

$$X[z] = \sum_{n=0}^{\infty} x[n]z^{-n} = x[0] + \frac{x[1]}{z} + \frac{x[2]}{z^2} + \frac{x[3]}{z^3} + \dots$$

Z Transform:

Example: $x[n] = \gamma^n u[n]$

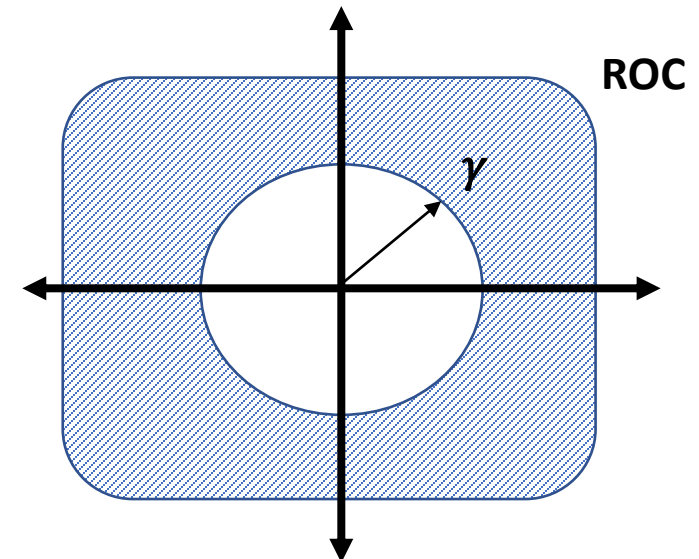
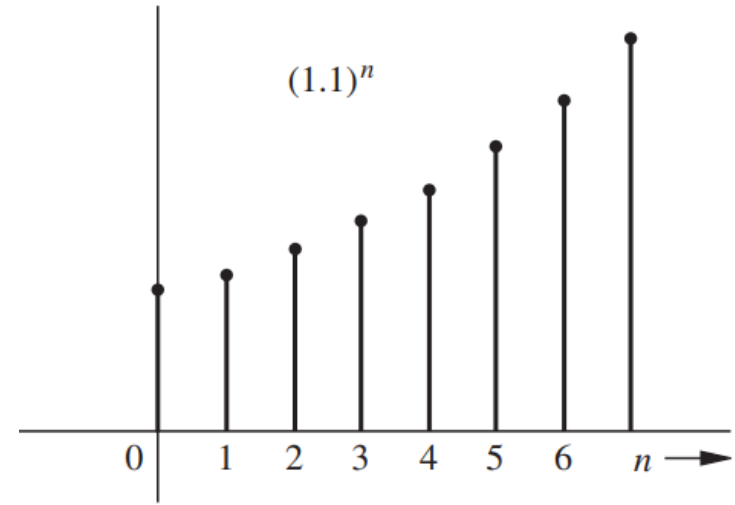
$$X(z) = \frac{z}{z - \gamma} \quad |z| > |\gamma|$$

If the signal grows like γ^n , then the weighting z^{-n} must decay faster to ensure convergence.

$$X[z] = \sum_{n=0}^{\infty} x[n]z^{-n} = x[0] + \frac{x[1]}{z} + \frac{x[2]}{z^2} + \frac{x[3]}{z^3} + \dots$$

↑
↑

ROC lies outside the circle $|z| = |\gamma|$.



Z Transform:

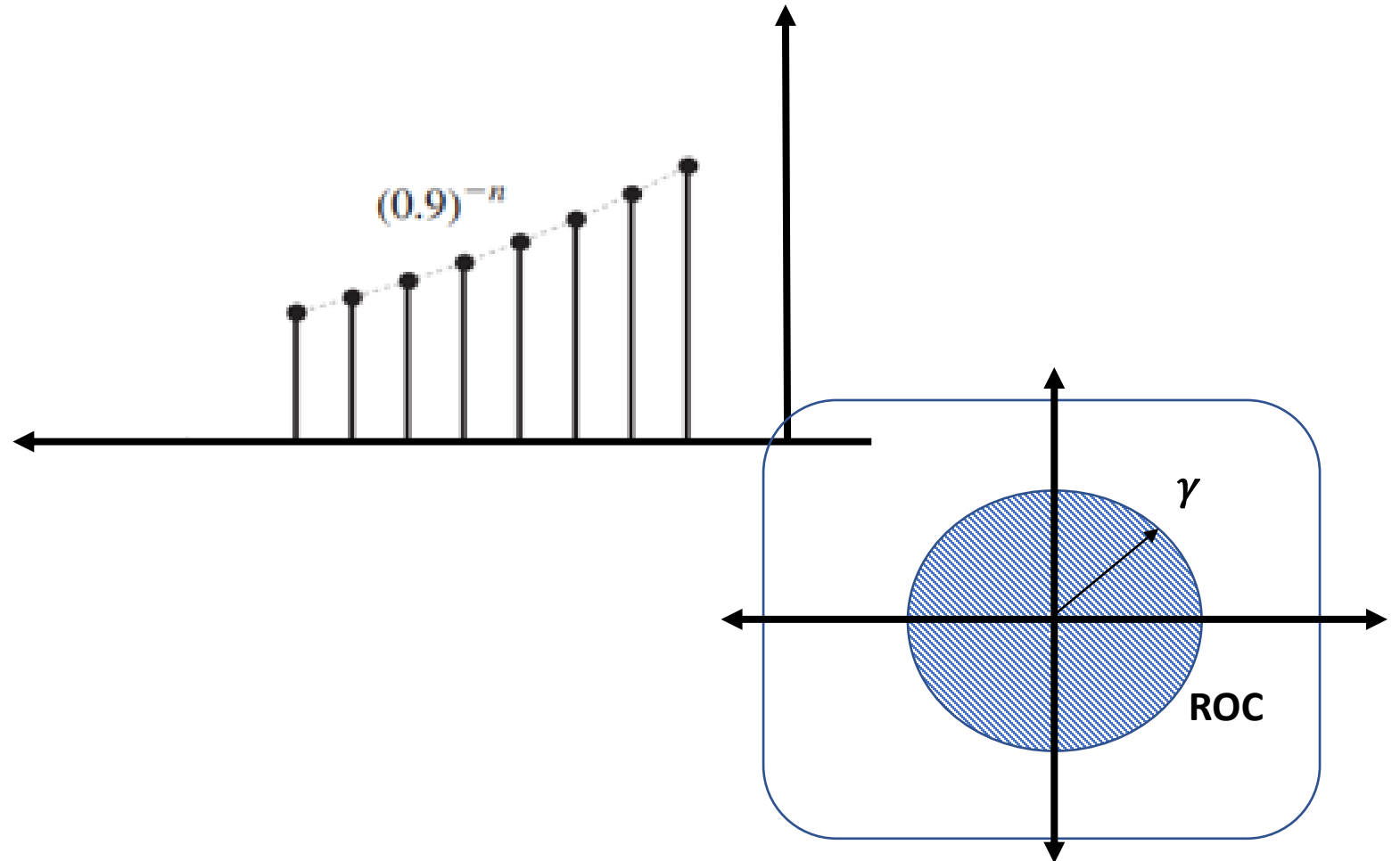
Example $x[n] = \gamma^n u[-n-1]$

$$X(z) = \sum_{n=-\infty}^{-1} (\gamma z^{-1})^n$$

$$X(z) = \frac{z}{\gamma - z}$$

$$\left| \frac{z}{\gamma} \right| < 1 \implies |z| < |\gamma|$$

ROC: $|z| < |\gamma|$

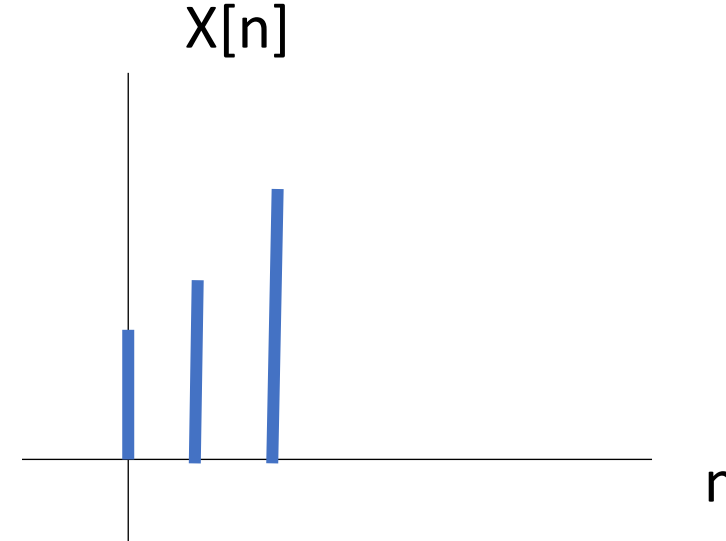


Z Transform:

Example: let $x[n] = [1, 2, 3]$.

Finite DT signal

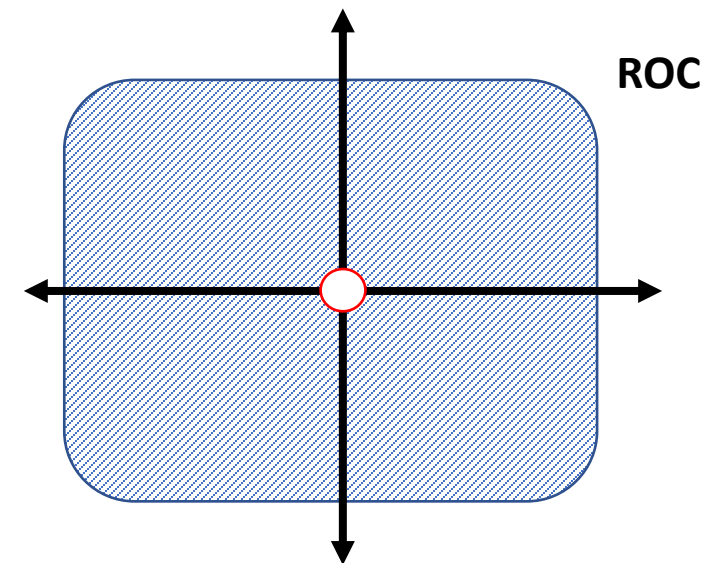
$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$



$$X(Z) = x[0]z^0 + x[1]z^{-1} + x[2]z^{-2} + x[3]z^{-3} + \dots$$

$$X(Z) = 1 z^0 + 2 z^{-1} + 3 z^{-2} + 0 z^{-3} + \dots$$

$$X(Z) = 1 + 2 z^{-1} + 3 z^{-2}$$



There is no infinite summation here; it's just a few terms. So convergence is not about a series.

ROC: entire z-plane except $z=0$

The center $z=0$ corresponds to extreme decay

Z Transform:

Example: let $x[n] = \delta[n]$ – a delta function.

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

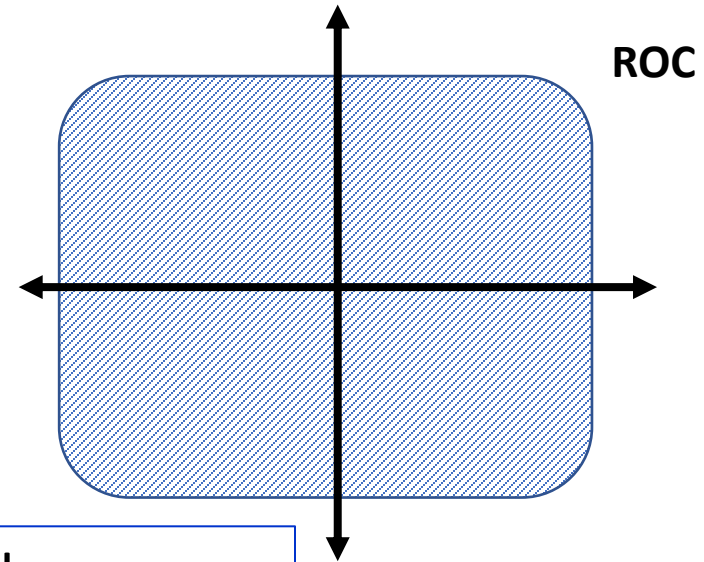
$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

$$X(z) = \sum_{n=0}^{\infty} \delta[n] z^{-n}$$

Since $\delta[n] = 0$ for $n \neq 0$, the sum reduces to:

$$X(z) = \delta[0] z^0 = 1$$

ROC: Entire Z-plane



Since the delta signal exists only at $n = 0$, it is completely unaffected by any exponential weighting z^{-n} , and every point in the Z-plane produces the same response. Therefore, the Z-transform of $\delta[n]$ is constant everywhere, and its ROC is the entire Z-plane.

Z Transform:

Example: let $x[n] = \delta[n - k]$ – a shifted delta function.

$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n - k] z^{-n}$$

Since the delta is nonzero only at $n=k$

$$X(z) = z^{-k}$$

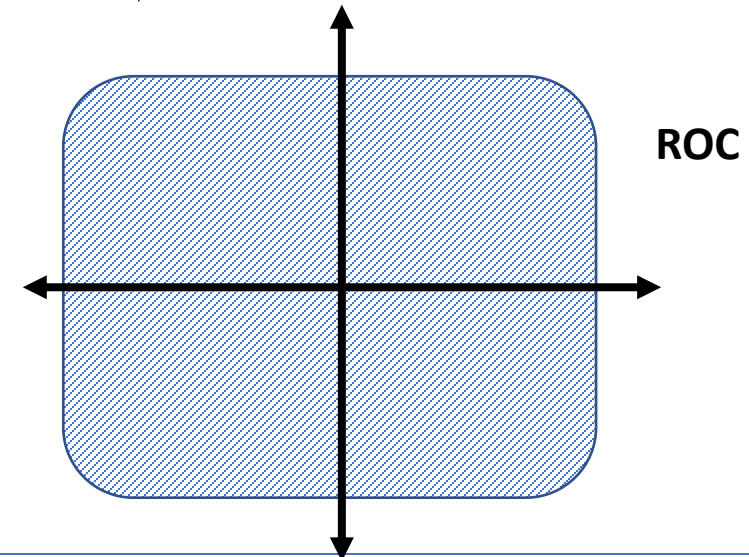
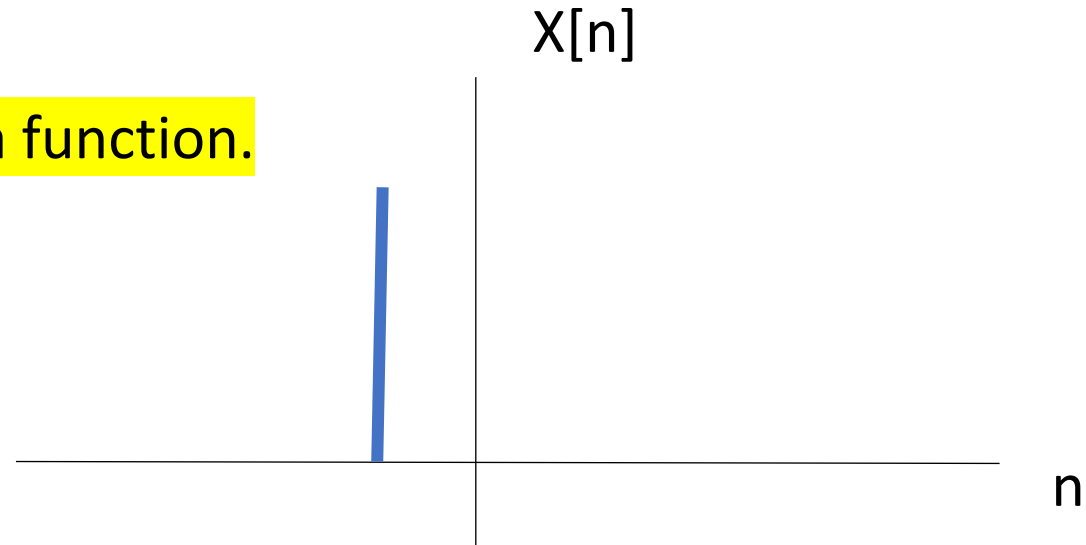
Case # 03 - When $K < 0$ - Right shift

$$X(Z) = Z^K$$

$X(Z)$ becomes infinite at

$$Z = \infty$$

Practically all finite z



ROC: entire z -plane except $z = \infty$

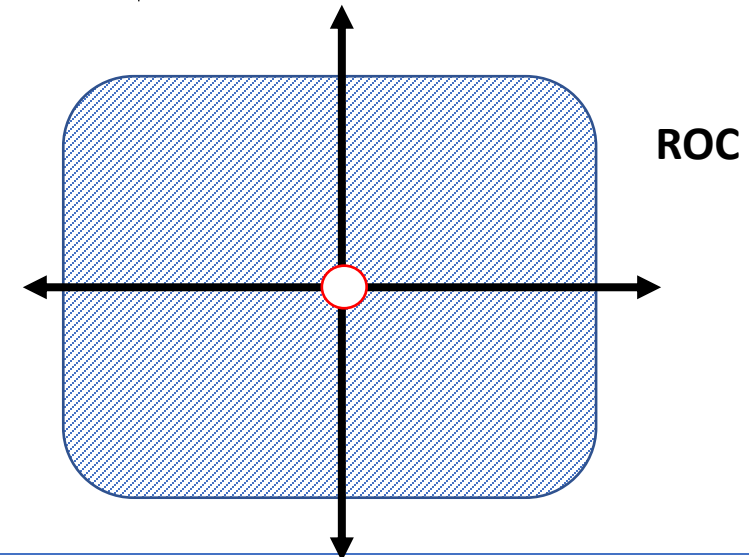
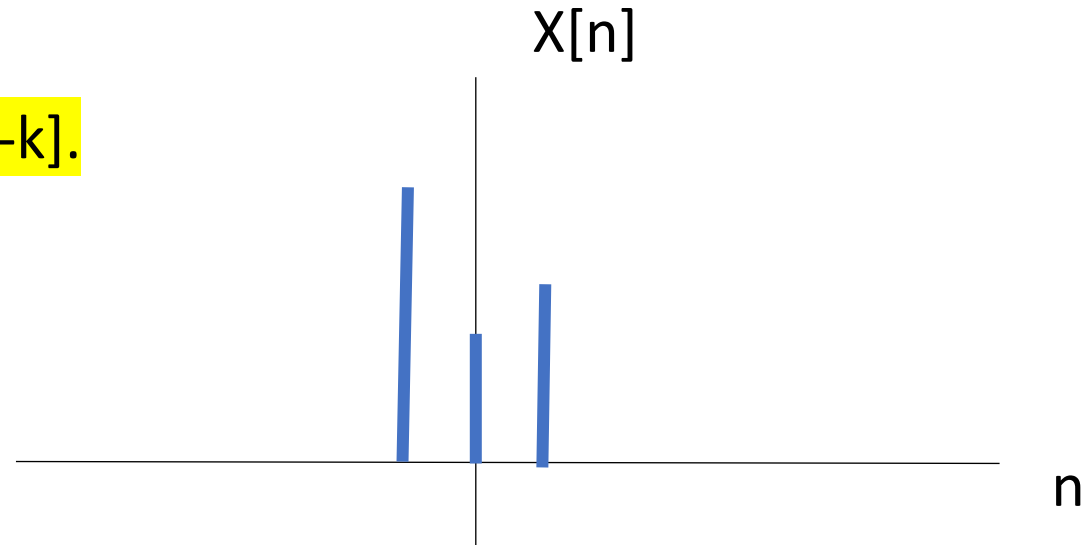
Z Transform:

Example: let $x[n] = 3\delta[n+k] + \delta[n] + 2\delta[n-k]$.

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$X(Z) = x[-1]z^1 + x[0]z^0 + x[1]z^{-1}$$

$$X(Z) = 3z^1 + 1z^0 + 2z^{-1}$$



ROC: entire z -plane except $0 < z < \infty$

Z Transform using properties:

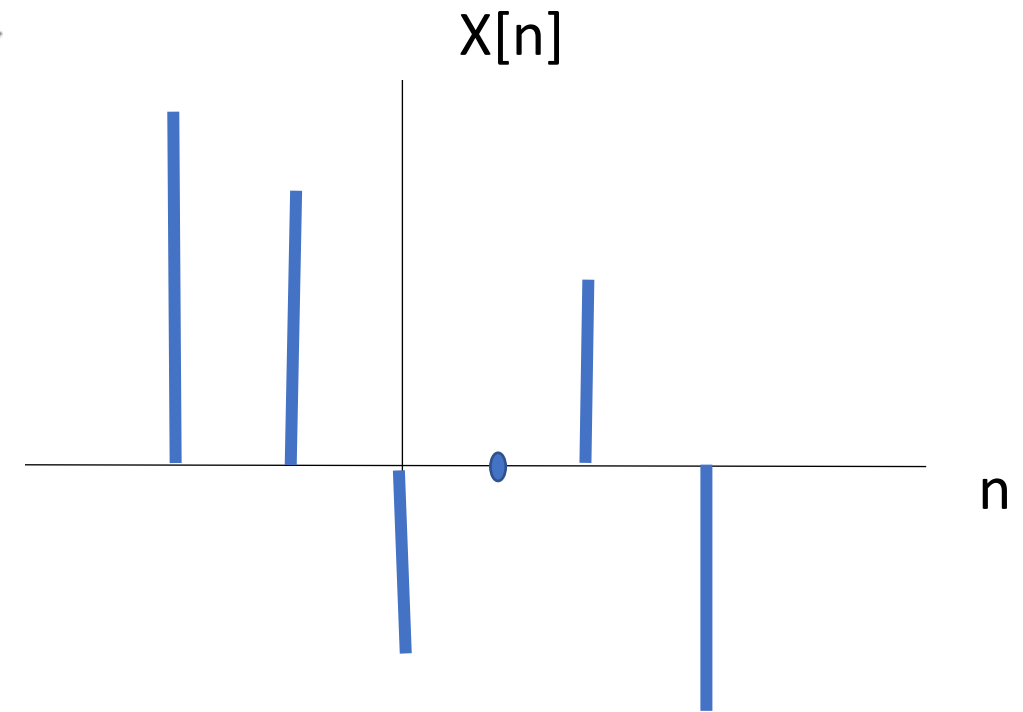
A finite sequence $x[n]$ is defined as

$$x[n] = \{5, 3, -2, 0, 4, -3\}$$

↑

Find $X(z)$ and its ROC.

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \sum_{n=-2}^3 x[n]z^{-n} \\ &= x[-2]z^2 + x[-1]z + x[0] + x[1]z^{-1} + x[2]z^{-2} + x[3]z^{-3} \\ &= 5z^2 + 3z - 2 + 4z^{-2} - 3z^{-3} \end{aligned}$$



$X(z)$ has a finite number of terms and consequently $X(z)$ will converge. As the $X(z)$ includes both positive powers of z and negative powers of z , the ROC of $X(z)$ is $0 < |z| < \infty$.



Z Transform using properties:

Find the z-transform $X(z)$ with the ROC for each of the following sequences:

$$(a) \quad x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[n]$$

$$(b) \quad x[n] = \left(\frac{1}{3}\right)^n u[n] + \left(\frac{1}{2}\right)^n u[-n-1]$$

$$(c) \quad x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[-n-1]$$



Z Transform Properties:

PROPERTY	SEQUENCE	TRANSFORM	ROC
	$x[n]$	$X(z)$	R
	$x_1[n]$	$X_1(z)$	R_1
	$x_2[n]$	$X_2(z)$	R_2
Linearity	$a_1x_1[n] + a_2x_2[n]$	$a_1X_1(z) + a_2X_2(z)$	$R' \supset R_1 \cap R_2$
Time shifting	$x[n - n_0]$	$z^{-n_0} X(z)$	$R' \supset R \cap \{0 < z < \infty\}$
Multiplication by z_0^n	$z_0^n x[n]$	$X\left(\frac{z}{z_0}\right)$	$R' = z_0 R$
Multiplication by $e^{j\Omega_0 n}$	$e^{j\Omega_0 n} x[n]$	$X(e^{-j\Omega_0} z)$	$R' = R$
Time reversal	$x[-n]$	$X\left(\frac{1}{z}\right)$	$R' = \frac{1}{R}$
Multiplication by n	$nx[n]$	$-z \frac{dX(z)}{dz}$	$R' = R$
Accumulation	$\sum_{k=-\infty}^n x[k]$	$\frac{1}{1-z^{-1}} X(z)$	$R' \supset R \cap \{ z > 1\}$
Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	$R' \supset R_1 \cap R_2$

$R' \supset R_1 \cap R_2$ - here R' (new ROC):
contains all elements that are common
to R_1 and R_2 – intersection.

Z Transform using properties:

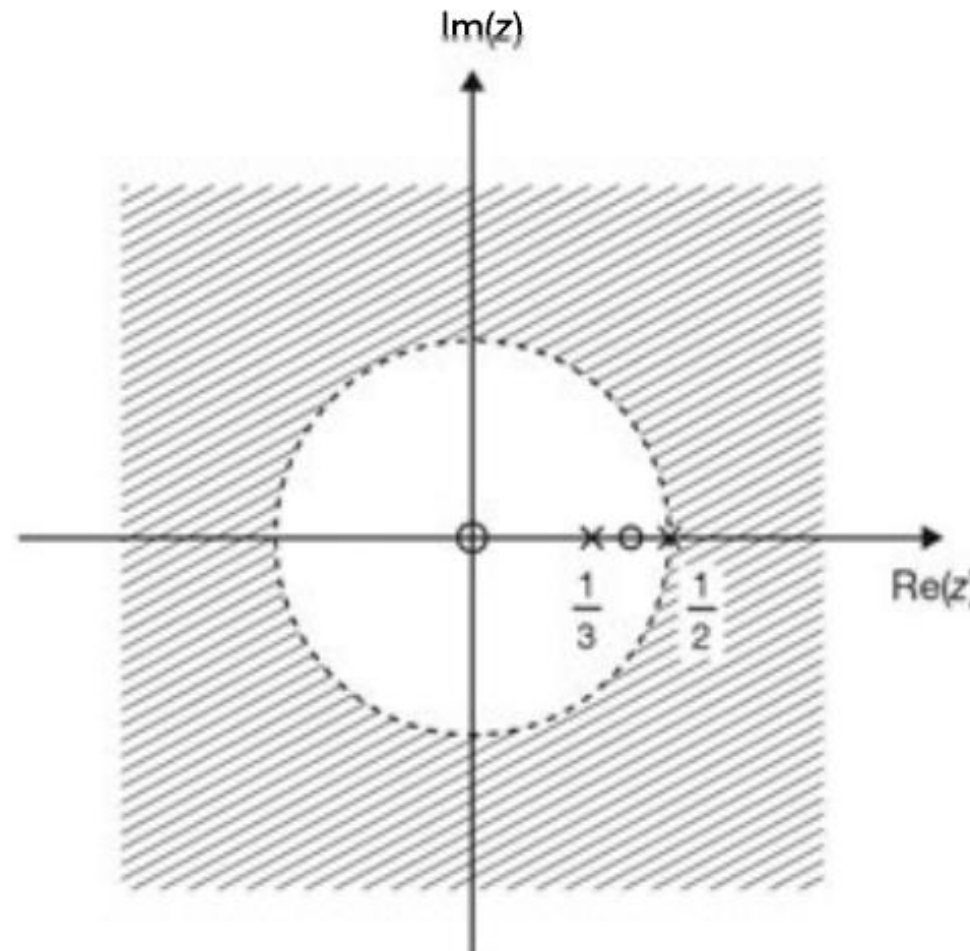
$$(a) \quad x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[n]$$

$$\text{ROC is } |z| > \frac{1}{2}$$

$$\left(\frac{1}{2}\right)^n u[n] \leftrightarrow \frac{z}{z - \frac{1}{2}} \quad |z| > \frac{1}{2}$$

$$\left(\frac{1}{3}\right)^n u[n] \leftrightarrow \frac{z}{z - \frac{1}{3}} \quad |z| > \frac{1}{3}$$

$$X(z) = \frac{z}{z - \frac{1}{2}} + \frac{z}{z - \frac{1}{3}}$$



Z Transform using properties:

$$(b) \quad x[n] = \left(\frac{1}{3}\right)^n u[n] + \left(\frac{1}{2}\right)^n u[-n-1]$$

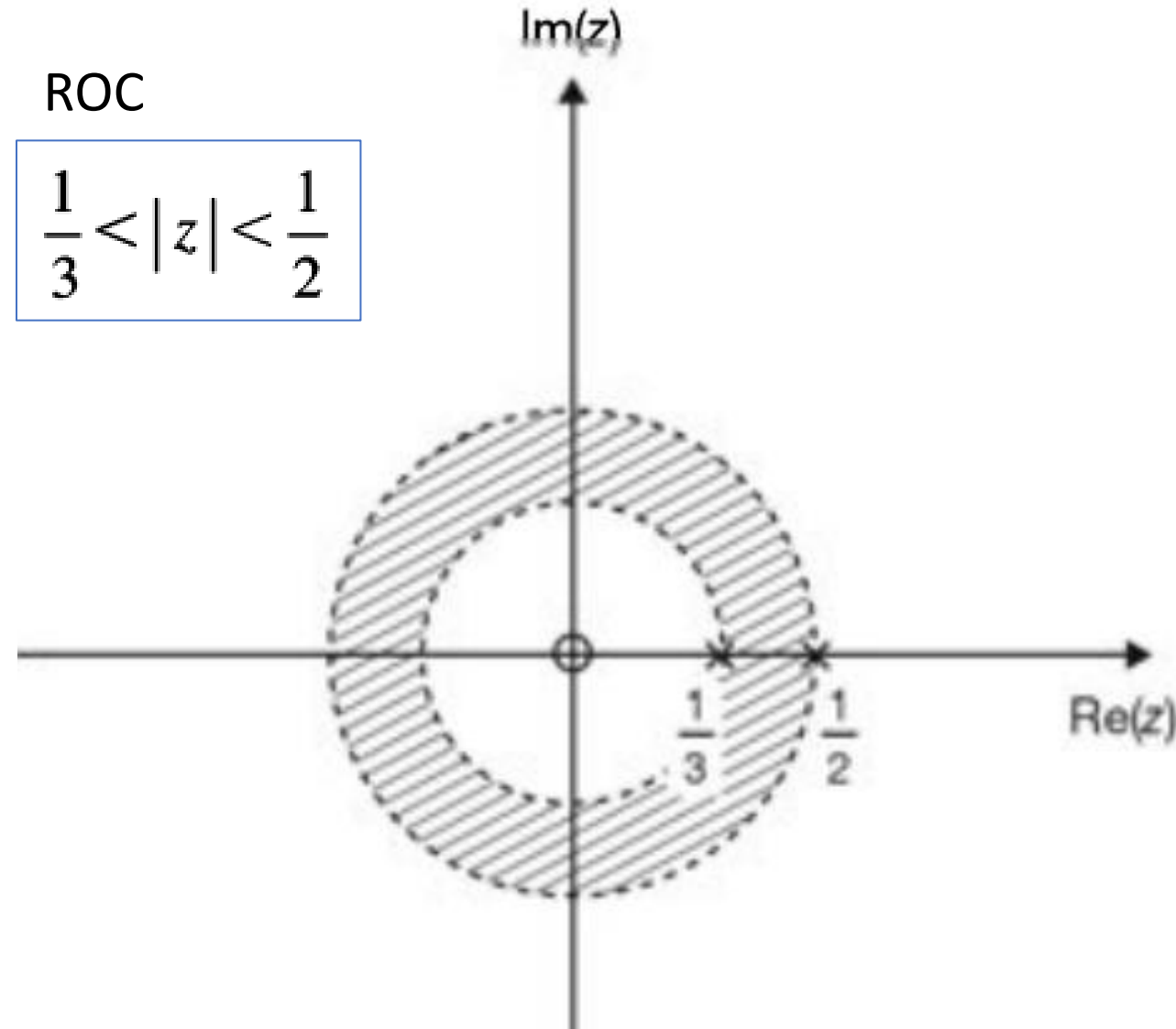
$$\left(\frac{1}{3}\right)^n u[n] \leftrightarrow \frac{z}{z - \frac{1}{3}} \quad |z| > \frac{1}{3}$$

$$\left(\frac{1}{2}\right)^n u[-n-1] \leftrightarrow -\frac{z}{z - \frac{1}{2}} \quad |z| < \frac{1}{2}$$

$$X(z) = \frac{z}{z - \frac{1}{3}} - \frac{z}{z - \frac{1}{2}}$$

ROC

$$\frac{1}{3} < |z| < \frac{1}{2}$$





Z Transform using properties:

$$(c) \quad x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[-n-1]$$

$$\left(\frac{1}{2}\right)^n u[n] \leftrightarrow \frac{z}{z - \frac{1}{2}} \quad |z| > \frac{1}{2}$$

$$\left(\frac{1}{3}\right)^n u[-n-1] \leftrightarrow -\frac{z}{z - \frac{1}{3}} \quad |z| < \frac{1}{3}$$

The ROCs do not overlap, and there is no common ROC; thus, $x[n]$ will not have $X(z)$.

Z Transform using properties:

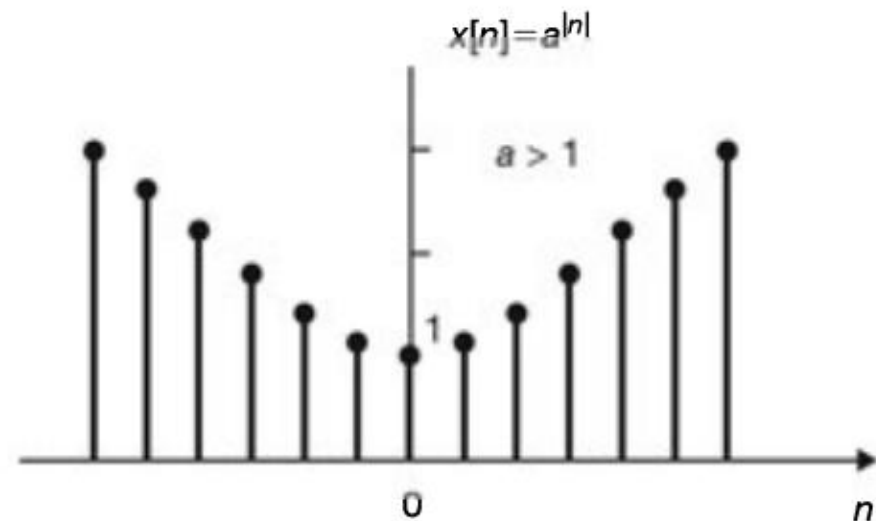
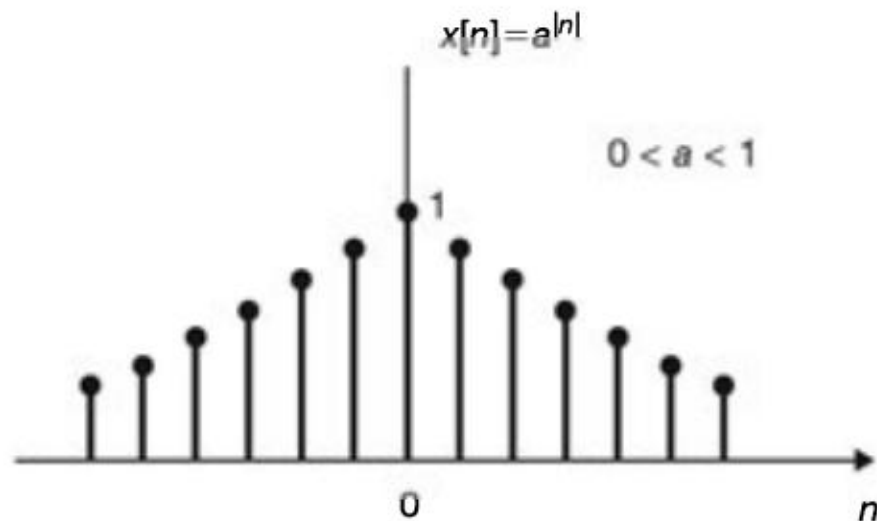
Let

$$x[n] = a^{|n|} \quad a > 0$$

(a) Sketch $x[n]$ for $a < 1$ and $a > 1$.

(b) Find $X(z)$ and sketch

the ROC for $a < 1$ and $a > 1$.



Z Transform using properties:

$$x[n] = a^n u[n] + a^{-n} u[-n - 1]$$

$$a^n u[n] \leftrightarrow \frac{z}{z - a} \quad |z| > a$$

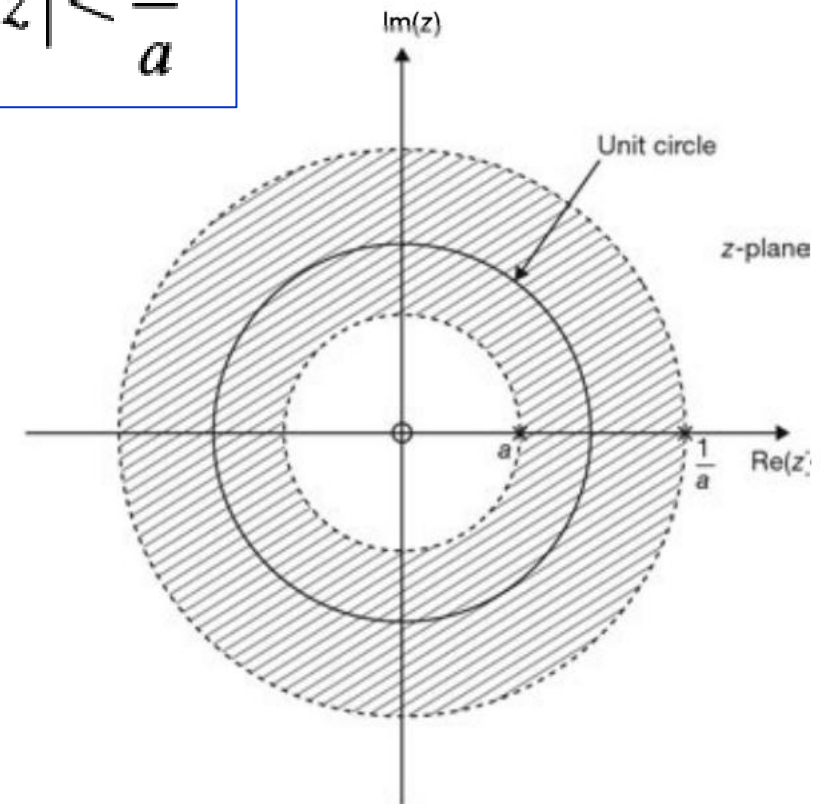
$$a^{-n} u[-n - 1] \leftrightarrow -\frac{z}{z - 1/a} \quad |z| < \frac{1}{a}$$

$$X(z) = \frac{z}{z - a} - \frac{z}{z - 1/a}$$

ROC

$$a < |z| < \frac{1}{a}$$

Overlap exists only for $a < 1$



The Z Transform

Inverse Z- Transform

Z Transform:

$$X[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$x[n] = \frac{1}{2\pi j} \oint X[z]z^{n-1} dz$$

$$X[z] = \mathcal{Z}\{x[n]\} \quad \text{and} \quad x[n] = \mathcal{Z}^{-1}\{X[z]\}$$

$$x[n] \Longleftrightarrow X[z]$$

How to get the inverse Z transform:

- I. MATLAB or any tool
- II. Partial fraction expansion – Z transform properties & Z transform table
- III. Long division – not a closed-form solution

The Z Transform

Partial fraction Expression



Z Transform:

Find the inverse z-transforms of $X[Z] = \frac{8z - 19}{(z - 2)(z - 3)}$

TABLE 5.1 Select (Unilateral) z-Transform Pairs

No.	$x[n]$	$X[z]$
1	$\delta[n - k]$	z^{-k}
2	$u[n]$	$\frac{z}{z - 1}$
3	$nu[n]$	$\frac{z}{(z - 1)^2}$
4	$n^2u[n]$	$\frac{z(z + 1)}{(z - 1)^3}$
5	$n^3u[n]$	$\frac{z(z^2 + 4z + 1)}{(z - 1)^4}$
6	$\gamma^n u[n]$	$\frac{z}{z - \gamma}$
7	$\gamma^{n-1} u[n - 1]$	$\frac{1}{z - \gamma}$
8	$n\gamma^n u[n]$	$\frac{\gamma z}{(z - \gamma)^2}$
9	$n^2\gamma^n u[n]$	$\frac{\gamma z(z + \gamma)}{(z - \gamma)^3}$

Use $x[n] \longleftrightarrow X(Z)$ lookup table to execute inverse Z transform

How to split the denominator to find the Z transform directly?

Technique- partial fraction expansion!



Z Transform:

Find the inverse z-transforms of $X[Z] = \frac{8z - 19}{(z - 2)(z - 3)}$

Partial fraction expansion by Cover-Up Method

The Cover-Up Method is a quick technique used to find coefficients in the partial fraction expansion of a proper rational function when the denominator has distinct linear factors.

$$\frac{N(z)}{(z - p_1)(z - p_2) \dots (z - p_n)} \quad \longrightarrow \quad \frac{A}{(z - p_1)} + \frac{B}{(z - p_2)} + \frac{C}{(z - p_3)} + \dots + \frac{N}{(z - p_n)}$$



Z Transform:

Find the inverse z-transforms of $X[Z] = \frac{8z - 19}{(z - 2)(z - 3)}$

Solution:

Partial fraction expansion:

$$\frac{8z - 19}{(z - 2)(z - 3)} = \frac{A}{z - 2} + \frac{B}{z - 3}$$

$A = (z - 2) X[Z] \big|_{z=2}$ Substitute $z = 2$ into the remaining factor $z - 3$:

$$A = \frac{8(2) - 19}{2 - 3} = \frac{16 - 19}{-1} = \frac{-3}{-1} = 3$$

$B = (z - 3) X[Z] \big|_{z=3}$ Substitute $z = 3$ into the remaining factor $z - 2$:

$$B = \frac{8(3) - 19}{3 - 2} = \frac{24 - 19}{1} = 5$$



Z Transform:

Find the inverse z-transforms of $X[Z] = \frac{8z - 19}{(z - 2)(z - 3)}$

Solution:

Partial fraction expansion:

$$\frac{8z - 19}{(z - 2)(z - 3)} = \frac{A}{z - 2} + \frac{B}{z - 3}$$

Expanding $X[z]$ into partial fractions yields:

$$X[z] = \frac{8z - 19}{(z - 2)(z - 3)} = \frac{3}{z - 2} + \frac{5}{z - 3} \quad \longrightarrow \quad x[n] = [3(2)^{n-1} + 5(3)^{n-1}]u[n - 1]$$

$$\gamma^{n-1}u[n - 1] \quad \frac{1}{z - \gamma}$$



Z Transform:

$$x[n] = [3(2)^{n-1} + 5(3)^{n-1}]u[n-1]$$

It prefers the form that contains $u[n]$ rather than $u[n-1]$. This is possible if the Z-transform has a “Z” term in the numerator.

This goal can be accomplished by expanding $X[z]/z$ into partial fractions instead of $X[z]$.

$\gamma^n u[n]$	$\frac{z}{z - \gamma}$
$\gamma^{n-1} u[n-1]$	$\frac{1}{z - \gamma}$

$$X[Z] = \frac{8z - 19}{(z - 2)(z - 3)} \quad \rightarrow \quad \frac{X[z]}{z} = \frac{8z - 19}{z(z - 2)(z - 3)}$$



Z Transform:

$$\frac{X[z]}{z} = \frac{8z - 19}{z(z-2)(z-3)}$$

$$= \frac{A}{z} + \frac{B}{(z-2)} + \frac{C}{(z-3)}$$

$$A = \frac{8(0) - 19}{(0-2)(0-3)} = \frac{-19}{(-2)(-3)} = -\frac{19}{6}$$

$$B = \frac{8(2) - 19}{2(2-3)} = \frac{16-19}{2(-1)} = \frac{-3}{-2} = \frac{3}{2}$$

$$C = \frac{8(3) - 19}{3(3-2)} = \frac{24-19}{3(1)} = \frac{5}{3}$$



Z Transform:

$$\frac{X[z]}{z} = \frac{8z - 19}{z(z-2)(z-3)}$$
$$= \frac{(-19/6)}{z} + \frac{(3/2)}{z-2} + \frac{(5/3)}{z-3}$$

Multiplying both sides by z yields

$$X[z] = -\frac{19}{6} + \frac{3}{2} \left(\frac{z}{z-2} \right) + \frac{5}{3} \left(\frac{z}{z-3} \right)$$
$$x[n] = -\frac{19}{6} \delta[n] + \left[\frac{3}{2} (2)^n + \frac{5}{3} (3)^n \right] u[n]$$

$$\delta[n-k] \quad z^{-k}$$

$$\gamma^n u[n] \quad \frac{z}{z-\gamma}$$



Z Transform:

Find $y[n]$

$$Y[z] = \frac{z(3z^2 - 9.5z + 10.5)}{(z - 0.5)(z^2 - 5z + 6)}$$

Solution:

$$\frac{Y[z]}{z} = \frac{3z^2 - 9.5z + 10.5}{(z - 0.5)(z - 2)(z - 3)}$$

$$\frac{3z^2 - 9.5z + 10.5}{(z - 0.5)(z - 2)(z - 3)} = \frac{A}{z - 0.5} + \frac{B}{z - 2} + \frac{C}{z - 3}$$

Z Transform:

Solution:

$$\frac{3z^2 - 9.5z + 10.5}{(z - 0.5)(z - 2)(z - 3)} = \frac{A}{z - 0.5} + \frac{B}{z - 2} + \frac{C}{z - 3}$$

$$A = \frac{3z^2 - 9.5z + 10.5}{(z - 2)(z - 3)} \quad z = 0.5$$

$$B = \frac{3z^2 - 9.5z + 10.5}{(z - 0.5)(z - 3)} \quad z = 2$$

$$C = \frac{3z^2 - 9.5z + 10.5}{(z - 0.5)(z - 2)} \quad z = 3$$

$$\frac{Y[z]}{z} = \frac{3z^2 - 9.5z + 10.5}{(z - 0.5)(z - 2)(z - 3)} = \frac{(26/15)}{z - 0.5} - \frac{(7/3)}{z - 2} + \frac{(18/5)}{z - 3}$$

$$Y[z] = \frac{26}{15} \left(\frac{z}{z - 0.5} \right) - \frac{7}{3} \left(\frac{z}{z - 2} \right) + \frac{18}{5} \left(\frac{z}{z - 3} \right)$$

$$y[n] = \left[\frac{26}{15} (0.5)^n - \frac{7}{3} (2)^n + \frac{18}{5} (3)^n \right] u[n]$$



Z Transform:

DRILL 5.2 Inverse z -Transform by Partial Fraction Expansion

Find the inverse z -transform of the following functions:

(a) $\frac{z(2z - 1)}{(z - 1)(z + 0.5)}$

(b) $\frac{1}{(z - 1)(z + 0.5)}$

ANSWERS

(a) $\left[\frac{2}{3} + \frac{4}{3}(-0.5)^n\right]u[n]$

(b) $-2\delta[n] + \left[\frac{2}{3} + \frac{4}{3}(-0.5)^n\right]u[n]$



Midterm Exam:

CLO-1	Apply knowledge of signals and systems to categorize and perform basic operations on signals and systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-2	Calculate the response of DT and CT LTI systems to various inputs and categorize LTI systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-3	Analyze DT signals and systems in the transform domain using the Z transform.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-4	Analyze CT signals and systems in the transform domain using Laplace, Fourier transforms, and Fourier Series.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-5	Investigate complex engineering problems in signals and systems and formally present the results.	PLO-4 (Investigation)	Cognitive Level-5 (Evaluating)



Thanks!